PKIaaS.io Documentation

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1. Getting Started

1.1 Login

Getting started with PKIaaS.io is easy! All you need to do is click "Manage PKI" at the top right of the main page and choose the login you'd like to use.

1.2 Create a Certificate Authority

All the certificates you'll create need to be signed by a certificate authority, so the first step is to create one. Navigating to any other page of the site before you've created a CA will redirect you back to a page forcing you to create one. If you're logging in for the first time, and you haven't created a CA before, you'll be asked to choose between a managed CA and a CA backed by an HSM. If you're new to PKI or you'd just like to take PKIaaS.io for a test drive, it would be best to choose the managed option to get going more quickly. Read more about these options here.

Note: PKIaaS.io is a comprehensive PKI platform that requires the creation of a certificate authority to issue certificates. However, if your use case involves generating a simple self-signed certificate without the need for a certificate authority, a straightforward tool like CertificateTools.com may be a more convenient alternative.

1.2.1 CA Configuration Options

If you're still learning about PKI, feel free to use the default options and just update the name when creating your first CA. This is the easiest option to start issuing certificates and getting a feel for the site. When you're done choosing options, click "Create New CA". You'll then be redirected to the "Manage CAs" page and your new CA will appear in the dropdown at the top of the page.

1.2.2 Switch CAs

If you create more than one CA, you'll have the option of switching between them by selecting the one you'd like to manage from the dropdown at the top of the PKI management page to the right of the navigation pane. All other pages on the site will show only the resources—templates, certificates, services, etc.—within the context of the selected CA.

1.3 Using Your New CA

Select the CA you want to work with from the dropdown at the top to the right of the navigation pane. This dropdown changes the scope of the site to manage the active CA. All actions performed—signing certificates, creating templates, creating ACME policies, etc.—will be done within the context of the selected CA.

1.4 Certificate Templates

1.4.1 Manage Templates

Certificate templates can be managed by navigating to **Certificate Templates -> Manage Templates** in the navigation pane on the left of the page. Certificate templates are used to enforce compliance of predefined standards for issuing certificates with your CA. These standards include key usages, subject attributes, basic constraints, and other certificate attributes. The use of templates is important to prevent accidental mis-issuance and to ensure that issued certificates are compliant with your security practices. All certificate signing on PKIaaS.io will be associated with a template.

1.4.2 Create New Templates

By default, a "Web Server" template is automatically built for every CA that is created. This template has a good out-of-the-box configuration for issuing TLS/SSL certificates for web servers, but new, custom certificate templates can be created by

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navigating to **Certificate Templates -> Create Template**. Enter a name for the template and choose default settings by selecting an option from the "Starter Template" dropdown. Click "Create Template" to proceed to the configuration page, where the template options can be customized. See more information on the available options here. Once satisfied with the settings, click "Save" to finalize the changes and return to the template management page.

1.5 Issue a Certificate

In the navigation pane on the left, navigate to **Certificate Templates -> Manage Templates** and click on the template you'd like to use. From the menu, you can choose either "Public Certificate Request", "Submit CSR", or "Create Pre-Approved Request." These options support various workflows for issuing certificates.

1.5.1 Public Certificate Request

If your workflow involves receiving certificate requests from users, the "Public Certificate Request" option allows you to send an email to users inviting them to easily submit a certificate request. It also shows the public URL that can be shared for others to submit a certificate request. This option allows users to submit a CSR or fill out a form containing the desired subject attributes and SANs. The latter option can be helpful for users that are not familiar with x509 certificates and includes the ability to generate an RSA private key automatically in the browser when the user claims their certificate after it is approved. To review and approve these requests, navigate to **X509 Certificates -> Pending Requests**.

1.5.2 Submit CSR

The "Submit CSR" option provides an input field to paste the contents of a certificate signing request. To review and approve these requests, navigate to **X509 Certificates -> Pending Requests**. After the request is approved, the submitted CSR will be signed and the certificate will appear in the **X509 Certificates -> Issued Certificates** page.

1.5.3 Create Pre-Approved Request

The "Create Pre-Approved Request" feature enables a PKI administrator to pre-configure all the settings typically chosen during the approval process for a pending certificate request. The administrator can either redeem the certificate immediately or send an email to the user, inviting them to claim it. In the certificate claim workflow, the user can set a password to protect the PKCS#12 file, and a 2048-bit RSA private key will be generated in the browser. Alternatively, a CSR can be submitted; however, the subject attributes and SANs specified in it will be overridden by the pre-approved attributes in the signed certificate.

1.6 Approve Pending Certificate Requests

Certificates requested using the "Public Certificate Request" or "Submit CSR" options require approval. The requests pending approval can be found by navigating to **X509 Certificates -> Pending Requests**. Here you will find options to Approve, Sign, Decline, or Resend emails for various types of certificate requests. Not all options are available for all types of certificate request workflows. These options can be performed in bulk, or the requests can be selected individually where additional options are provided to override subject attributes, subject alternative names (SANs), and to customize the validity time of the certificate instead of taking the template default.

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2. Patreon Subscription

PKIaaS.io relies on community support. By subscribing on Patreon, you help sustain the platform's development and maintenance. Patreon subscribers enjoy lifted account restrictions and direct communication with support.

To subscribe, log in to PKIaaS.io, click on your account name in the top right corner, and select "Profile." From there, click the "Become a Patron" link to log in to Patreon and choose a subscription tier.

Unsubscribed users are subject to the following limitations:

- Maximum of 10 certificates per account (excluding service certificates, e.g., SCEP, OCSP)
- 30-day certificate validity period
- No certificate revocation

If a certificate authority is delegated to other users, only the CA owner must be a Patreon subscriber to lift the limitations for all delegated users using the CA. If an unsubscribed user delegates a CA to a user with an active Patreon subscription, the limitations will apply to the subscribed users using the CA.

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3. Post-Quantum Cryptography

3.1 Overview

Post-quantum cryptography (PQC) refers to cryptographic algorithms designed to remain secure against the potential threats posed by quantum computers. Quantum computers have the potential to break many of the cryptographic algorithms that are currently in use, including RSA and ECC. As a result, the National Institute of Standards and Technology (NIST) has been working to standardize post-quantum cryptographic algorithms.

On August 13, 2024, NIST finalized two digital signature standards that can be used to create quantum-safe X.509 certificates: ML-DSA (CRYSTALS-Dilithium), as defined in FIPS 204, and SLH-DSA (formerly SPHINCS+), as defined in FIPS 205. PKIaaS.io currently supports ML-DSA and SLH-DSA.

- ML-DSA
- SLH-DSA

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3.2 ML-DSA

3.2.1 Create an ML-DSA Certificate Authority

Log in to PKIaaS.io, and navigate to **Certificate Authorities -> Create New CA**. From the **Key Type** dropdown, select "ML-DSA". In the **Parameter Set** dropdown, "65" (security level 3) will be selected by default. A different parameter set may be selected if desired to customize the security level of the private key for the CA. See more information about ML-DSA security levels below. Refer to Certificate Authorities Overview for more information on the available options in the new CA form. After the form has been completed, click **Create New CA** to create the ML-DSA CA.

Security Levels (Parameter Sets)

ML-DSA offers parameter sets that meet three security levels: ML-DSA-44, ML-DSA-65, and ML-DSA-87. ML-DSA-44 is intended to meet NIST's security level 2, ML-DSA-65 is intended to meet level 3, and ML-DSA-87 is intended to meet level 5. See https://www.ietf.org/archive/id/draft-salter-lamps-cms-ml-dsa-00.html for more information.

- **Security Level 2 (ML-DSA-44)** - 160-bit AES equivalent and 4096-bit RSA equivalent
- **Security Level 3 (ML-DSA-65)** - 192-bit AES equivalent and 7680-bit RSA equivalent
- **Security Level 5 (ML-DSA-87)** - 256-bit AES equivalent and 15360-bit RSA equivalent

3.2.2 Generate ML-DSA Private Keys and CSRs

In order to issue an ML-DSA certificate, a CSR must be generated with an ML-DSA private key. This can be done using tools like OpenSSL (version 3.5 or higher) or online at https://certificatetools.com/ by selecting one of the ML-DSA private key options from the **Private Key** dropdown.

Note: In production, it is recommended to generate the private key and CSR on a secure system that is not connected to the internet.

3.2.3 Issue an ML-DSA Certificate

Navigate to **Certificate Templates -> Manage Templates**, and click on the desired template. From the menu select **Submit CSR**. Paste the contents of a certificate signing request (CSR) generated with an ML-DSA private key and click **Submit**. Navigate to **X509 Certificates -> Pending Requests** to review and approve the request. Click on the pending request to view the details, optionally override requested attributes or validity, and click **Sign** to issue the certificate. Navigate to **X509 Certificates -> Issued Certificates** to view and download the issued certificate.

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3.3 SLH-DSA

3.3.1 Create an SLH-DSA Certificate Authority

Log in to PKIaaS.io, and navigate to **Certificate Authorities -> Create New CA**. From the **Key Type** dropdown, select "SLH-DSA". Select the desired parameter set from the **Parameter Set** dropdown. See more information about SLH-DSA security levels below. Refer to Certificate Authorities Overview for more information on the available options in the new CA form. After the form has been completed, click **Create New CA** to create the SLH-DSA CA.

Security Levels (Parameter Sets)

SLH-DSA offers 12 parameter sets that meet three security levels. SLH-DSA-SHA2-128s, SLH-DSA-SHA2-128f, SLH-DSA-SHAKE-128s and SLH-DSA-SHAKE-128f are intended to meet NIST's level 1 security level. SLH-DSA-SHA2-192s, SLH-DSA-SHA2-192f, SLH-DSA-SHAKE-192s and SLH-DSA-SHAKE-192f are intended to meet level 3. SLH-DSA-SHA2-256s, SLH-DSA-SHA2-256f, SLH-DSA-SHAKE-256s and SLH-DSA-SHAKE-256f are intended to meet level 5. See https://www.ietf.org/archive/id/draft-ietf-lamps-x509-slhdsa-01.html for more information.

- **Security Level 1 (SLH-DSA-SHA2|SHAKE-128s|f)** - 128-bit AES equivalent and 3072-bit RSA equivalent
- **Security Level 3 (SLH-DSA-SHA2|SHAKE-192s|f)** - 192-bit AES equivalent and 7680-bit RSA equivalent
- **Security Level 5 (SLH-DSA-SHA2|SHAKE-256s|f)** - 256-bit AES equivalent and 15360-bit RSA equivalent

3.3.2 Generate SLH-DSA Private Keys and CSRs

In order to issue an SLH-DSA certificate, a CSR must be generated with an SLH-DSA private key. This can be done using tools like OpenSSL (version 3.5 or higher) or online at https://certificatetools.com/ by selecting one of the SLH-DSA private key options from the **Private Key** dropdown.

Note: In production, it is recommended to generate the private key and CSR on a secure system that is not connected to the internet.

3.3.3 Issue an SLH-DSA Certificate

Navigate to **Certificate Templates -> Manage Templates**, and click on the desired template. From the menu select **Submit CSR**. Paste the contents of a certificate signing request (CSR) generated with an SLH-DSA private key and click **Submit**. Navigate to **X509 Certificates -> Pending Requests** to review and approve the request. Click on the pending request to view the details, optionally override requested attributes or validity, and click **Sign** to issue the certificate. Navigate to **X509 Certificates -> Issued Certificates** to view and download the issued certificate.

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4. Certificate Authorities

4.1 Overview

A certificate authority (CA) serves as the foundation upon which all functionality on PKIaaS.io is built. Tasks like working with templates, issuing certificates, and managing ACME policies all operate within the scope of a CA. As such, the first step after signing into PKIaaS.io is creating a CA. Users will be automatically redirected to a setup page and prompted to create a CA before accessing other features of the site.

Users can choose between creating a managed CA or one backed by an HSM. With a managed CA, the private key is securely stored using AES-256 encryption, offering a streamlined setup without additional configuration. Alternatively, the HSM-backed option integrates with a PKIaaS.io IoT-HSM, enabling key management via a YubiKey 5. In this setup, all signing operations are routed to the IoT-HSM, where they are processed by the YubiKey. For detailed guidance on creating a CA with IoT-HSM, refer to the IoT-HSM documentation.

4.1.1 Options

Subject Attributes

When creating a certificate authority (CA), users are provided with a range of flexible options to tailor their setup. The first six fields customize the subject attributes of the CA, with only the "Certificate Authority Name" (common name) being mandatory.

Private Key

Next, users can configure the private key settings. Various private key types are supported including RSA, ECC, and CRYSTALS-Dilithium. Here, users can also opt to use HSM (as previously mentioned) or import an existing private key.

Hash Algorithm, Validity, and Path Length

Further customization options include selecting the signature hash algorithm, specifying the path length, and setting the certificate authority's validity period. The signature algorithm chosen here determines how the CA's certificate is self-signed. Depending on the private key used, additional signature options may become available for signing certificates within the certificate templates after the CA is established. For more details on "Path Length," refer to RFC 5280, which explains the "basic constraints" X.509v3 attributes.

Key Usage

It is not recommended to make changes to the default "Key Usage" and "Extended Key Usage" options. Making changes could result in issues with your certificate authority.

Intermediate CA

Finally, the "CSR Only" option is designed for cases where the CA being created will function as an intermediate or subordinate certificate authority rather than a root CA. This option generates a Certificate Signing Request (CSR) that must be signed by another CA. The signing can be performed by an external CA (such as Microsoft or EJBCA) or another PKIaaS.io CA. Once the signed certificate is obtained, it must be imported before the CA can be activated and used.

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4.2 Create Certificate Authorities

4.2.1 Create a Root CA

Users can create a root certificate authority by logging in to PKIaaS.io and navigating to **Certificate Authorities -> Create New CA**. Fill out the form with the appropriate options for the desired CA. See certificate authority overview for documentation on all of the available options. When creating a root CA, it is necessary to leave the "CSR Only" option at the bottom of the form unchecked. Upon clicking "Create New CA," a new private key will be generated, and a self-signed, root certificate authority will be created and ready to issue certificates. See the getting started guide for more documentation about issuing certificates.

4.2.2 Create an Intermediate/Subordinate CA

To create an intermediate/subordinate certificate authority, ensure that the "CSR Only" option is selected on the **Certificate Authorities -> Create New CA** page when selecting the options for the desired CA. See certificate authority overview for documentation on all of the available options. Upon clicking "Create New CA," a new private key will be generated, and a CSR will be created. The browser will be redirected to a page containing the CSR where it can be copied to the clipboard and signed by another certificate authority.

Note: If at any time you need to navigate back to the CSR page, go to **Certificate Authorities -> Manage CAs**, click on the CA and select "Download/Copy CSR" from the menu.

Sign the CSR with a Root CA on PKIaaS.io

If a certificate authority not managed by PKIaaS.io will be used to sign the CSR, see the documentation for the CA on how to do so. The following instructions will detail how this can be done on a CA managed by PKIaaS.io.

While logged into PKIaaS.io, select the signing root CA from the dropdown at the top of the page. This will switch the context of the admin page to the selected CA. If a template for creating certificate authorities has not already been created, navigate to **Certificate Templates -> Create Template**. Enter "Intermediate CA" as the template name and choose "Certificate Authority" as the starter template. The browser will be directed to a page where the template can optionally be customized. Click "Save" to finish creating the new template.

The browser should now be at the **Certificate Templates -> Manage Template** page. Click on the "Intermediate CA" template and select "Submit CSR" from the menu. Paste the CSR copied to the clipboard in a previous step and click submit. See the note above for instructions to navigate back to the CSR if needed.

The submitted CSR must now be approved in order to be signed. Navigate to **X509/Certificates -> Pending Requests**. Click on the pending certificate request for the new intermediate CA. A modal will appear where some certificate attributes can optionally be overridden. Click "Sign" to sign the submitted CSR and issue the certificate.

Finally, navigate to **X509/Certificates -> Issued Certificates**. Click on the newly issued certificate and copy it to the clipboard. This certificate must be imported to finish creating the new intermediate CA. Navigate to **Certificate Authorities -> Manage CAs**, click on the new intermediate CA and choose "Import Signed Certificate" from the menu. Paste the signed certificate into the text area and click "Import Certificate." The intermediate CA is now ready to issue certificates. Select it from the dropdown at the top of the page to begin using it.

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4.3 Edit Certificate Authorities

To edit a certificate authority, log in to PKIaaS.io and navigate to **Certificate Authorities -> Manage CAs** in the navigation pane on the left of the page. Click on a CA and select "Edit CA" from the menu. The CPS (certificate practice statement) can be customized here as well as the service certificate settings. The service certificate settings manage the key type, key size and hash algorithm used when creating certificates for services like timestamping and OCSP.

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4.4 Delete Certificate Authorities

To delete a certificate authority, log in to PKIaaS.io and navigate to **Certificate Authorities -> Manage CAs** in the navigation pane on the left of the page. Click on a CA and select "Edit CA" from the menu. Click the red "Delete" button to delete the CA. This action is not reversible and will delete all certificates and templates associated with the CA.

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4.5 Delegate CA Access

To allow multiple users to share management of a certificate authority (CA) on PKIaaS.io, you can delegate access to the CA. Delegation enables other users to issue certificates, manage templates, and perform other administrative tasks without needing to share account credentials.

4.5.1 Delegating a CA

To delegate access to a CA, follow these steps:

1. Log in to PKIaaS.io.
2. Navigate to **Certificate Authorities -> Manage CAs** in the left navigation pane.
3. Click on the CA you want to delegate.
4. Select **Delegate Access** from the menu.
5. On the delegation page, click "Invite".
6. Enter the email address of the user you want to delegate access to. You may optionally choose an authentication provider to force the user to log in with while logging in to accept the invitation and whether to require the login email to match the email address provided.
7. Click "Send Invitation" to send the invitation email to the user.
8. The invited user will receive an email with a link to accept the invitation. Once accepted, the user will have access to the delegated CA. The invitation must be accepted within 7 days, or it will expire.

> **Note:** When sending a delegation invitation, use of the "Authentication Provider" option and "Require login email to match invitation email" is recommended to improve security. Without these options, any user with access to the link in the invitation email can accept the invitation.

4.5.2 Resending Invitations

If the invited user does not receive the invitation email or it expires, the invitation can be resent. To resend an invitation:

1. Log in to PKIaaS.io.
2. Navigate to **Certificate Authorities -> Manage CAs** in the left navigation pane.
3. Click on the CA you want to resend the invitation for.
4. Select **Delegate Access** from the menu.
5. On the delegation page, select the user you want to resend the invitation to.
6. Click the "Resend" button.

4.5.3 Remove Delegated Access

To remove delegated access for a user:

1. Log in to PKIaaS.io.
2. Navigate to **Certificate Authorities -> Manage CAs** in the left navigation pane.
3. Click on the CA you want to remove delegated access from.
4. Select **Delegate Access** from the menu.
5. On the delegation page, select the user you want to remove.
6. Click the "Remove" button.

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4.5.4 Limitations

- The owner of the CA must be a Patreon subscriber to lift account limitations for all delegated users.
- PKIaaS.io currently does not support RBAC (Role-Based Access Control) for delegated CAs. All delegated users will have full administrative access to the CA, including the ability to delete it and delegate to other users.
- Delegated users may remove access to other delegated users, except the CA owner.

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5. Certificate Templates

5.1 Overview

Certificate templates are used to enforce compliance of predefined standards for issuing certificates with your certificate authority. These standards include key usages, subject attributes, basic constraints, and other certificate attributes. Certificate templates also serve as the entry point for many of the certificate issuance options on PKIaaS.io. Read more about these certificate issuance options on the getting started page.

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5.2 Create Templates

To create a new certificate template, log in to PKIaaS.io and navigate to **Certificate Templates -> Create Template**. Enter a name for the template and choose default settings by selecting an option from the "Starter Template" dropdown. Click "Create Template" to proceed to the configuration page, where the template options can be customized. Once satisfied with the settings, click "Save" to finalize the changes and return to the template management page.

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5.3 Edit Templates

To edit the settings for a certificate template, log in to PKIaaS.io and navigate to **Certificate Templates -> Manage Templates** in the navigation pane on the left of the page. Click on a template and select "Edit Template".

5.3.1 Options

General

- **Enabled** - enables/disables the template
- **Allow unsolicited certificate web requests** - enables/disables the "Public Certificate Request" URL for the template
- **Allow the use of a valid certificate to authorize an automatic renewal request** - enables/disables certificate renewal authorization using a previously issued certificate using the renewal options found on the CA's CPS page
- **Automatically revoke old certificate** - revoke the old certificate used to authorize renewal
- **Require certificate to be within** - don't allow renewal using a certificate to authorize renewal unless it is within the specified number of days of expiration
- **Copy subject and SANs from original certificate during renewal** - use the same subject and subject alternative names from the old certificate used to authorize renewal
- **Signing Algorithm** - customize the hash algorithm used to sign certificates
- **Validity Period (Days)** - specify the default length of time the issued certificate will be valid—for manually issued certificates. This option can be overridden at the time of signing when the request is in **X509 Certificates -> Pending Requests**
- **Description** - provide a description for the template

Subject

To honor the subject attributes requested within the CSR, use the **Supplied by the CSR** option. Otherwise, the subject attributes in the CSR will be overridden by the defaults defined here.

Key Usage

Select the key usages and extended key usages to be enabled for certificates issued using this template. Custom comma-separated extended key usage OIDs can be entered into the **Custom EKU OIDs** field.

Basic Constraints

The Basic Constraints extension in an X509 certificate defines whether the certificate is a Certificate Authority (CA) or an end-entity (non-CA) certificate. Set the **Certificate Authority** option to yes if the template will be used to sign subordinate (intermediate) certificate authorities. Path length defines the maximum number of subordinate CAs that can exist below this certificate in the certificate hierarchy.

CRL / OCSP / AIA / CPS

- **Include AIA Extension in Certificate** - The Authority Information Access (AIA) attribute is an optional but commonly included extension in an X.509 certificate. It provides information about how to access important resources or services related to the certificate issuer. Use this option to enable or disable the AIA attribute in certificates issued using this template.
- **Custom URL** - Allows customization of the URL where the issuing CA’s certificate (or chain) can be downloaded.
- **Include OCSP Extension in Certificate** - Enable or disable the OCSP Responder attribute within the AIA extension in certificates issued using this template.
- **Include CRL Extension in Certificate** - Enable or disable the CRL Distribution Point (CDP) attribute in certificates issued using this template.

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- **Include CPS Extension in Certificate** - Enable or disable the Certificate Policies attribute in certificates issued using this template. This attribute contains the URL to the Certificate Practice Statement (CPS) and the policy identifier.
- **Policy** - Disable, select a predefined policy identifier, or enter a custom OID.

SCEP

Simple Certificate Enrollment Protocol is a protocol designed to simplify the process of managing and issuing digital certificates in a scalable and automated manner. It is commonly used in environments where devices like routers, printers, and mobile devices need to enroll for certificates to enable secure communications. The SCEP URL for the template can be found by navigating to **Certificate Templates -> Manage Templates**, next click on a template, and choose the **Show Template Service URLs** option. A modal will appear revealing the template service URLs including the SCEP URL.

- **Enable SCEP protocol** - Enable or disable SCEP for the template.
- **Enable SCEP Static Passphrase** - Enable or disable a static passphrase for SCEP services on the template. This may be required for SCEP processes that require the same passphrase to be used for multiple SCEP clients.
- **Enable Microsoft Intune/Endpoint Manager Integration** - Enable or disable Microsoft Intune SCEP integration for the template, either connected automatically or with your own app registration. See Microsoft Intune Integration.
- **Strip Root from GetCACert Requests** - Exclude the root CA from responses to SCEP GetCACert requests, returning only the issuing CA and any intermediates.
- **Enable Microsoft SCEP compatibility** - Enable or disable the Microsoft SCEP (NDES) compatibility. This option enables PKIaaS.io to mimic the /CertSrv/mscep_admin/ URL on Microsoft NDES servers to generate "enrollment challenge passwords" for processes and workflows designed to work with this Microsoft NDES functionality. The URL for this page can be found by navigating to **Certificate Templates -> Manage Templates**, next click on a template, and choose the **SCEP Admin (Microsoft compatibility)** option to navigate to the URL in a new tab.
- **Enable Microsoft SCEP basic auth** - Enable or disable basic HTTP authentication for the Microsoft SCEP compatibility page. A username and password must be provided if this option is selected.
- **Enable IP ACL** - Enable or disable the IP-based access control list (firewall functionality) for access to SCEP requests, the Microsoft compatibility page (if enabled), or both. Enter a list of IPs allowed to access the resources.

CT Logs

For templates associated with root CAs or intermediate CAs that chain up to root CAs that require issued certificates to be submitted to certificate transparency logs, enter a comma-separated list of URLs where these certificates should be submitted here. The CT log must support logging for the CA associated with the template. This is something that must be coordinated with the owner of the CT log.

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5.4 Delete Templates

To delete a certificate template, log in to PKIaaS.io and navigate to **Certificate Templates -> Manage Templates** in the navigation pane on the left of the page. Click on a template and select "Edit Template". If the template has no active certificates, you can use the red Delete button to remove it. Note that all certificates issued under the template must either be expired or revoked, and the revocation reason cannot be "Certificate Hold".

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5.5 Service URLs

The SCEP services URL, ACME directory URL, and a URL to allow users to make public certificate requests can all be found by navigating to **Certificate Templates -> Manage Templates** while logged into PKIaaS.io, and then by clicking one of the templates and selecting the **Show Template Service URLs** option.

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6. Certificate Management

6.1 Pending Requests

"Pending Requests" is a list of certificate requests that have been submitted but not yet signed. To view the list of pending requests, log in to PKIaaS.io, and navigate to **X509 Certificates -> Pending Requests**. Click on the pending certificate request to view the details of the request. A modal will appear with the details of the request, including the CSR, the requested attributes, and the amount of time the certificate will be valid for, based on the template that was used to submit the request. At the bottom of the modal, you can approve or reject the request by clicking "Sign" or "Decline" respectively. If you approve the request, the certificate will be signed and the certificate will appear in the list of issued certificates by navigating to **X509 Certificates -> Issued Certificates**. See more information below for options to override the default validity period and attributes requested in the CSR.

6.1.1 Override Requested Attributes

Before approving a pending certificate request, you have the option to override the attributes requested in the CSR before it is signed. To override the requested attributes, click on the "Requested Attributes" tab in the pending request approval modal and click "Override requested attributes". The form will become active, allowing the existing attributes to be removed and new attributes to be added. This can be useful for adding missing SANs or subject attributes to the certificate.

6.1.2 Override Validity Period

Before approving a pending certificate request, you have the option to override the default validity period of the template that was used to submit the CSR. To override the validity period, click on the "Validity" tab in the pending request approval modal and click "Override validity time". The form will become active, allowing the validity period to be changed. Two options exist, allowing the administrator to choose whether to set the validity period in days relative to the current time or select the start and end date and time of validity. The latter option can be used to ensure the certificate isn't valid until a specific date in the future.

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6.2 Revoke Certificates

6.2.1 Revoke a Certificate

To revoke a certificate, log in to PKIaaS.io and navigate to **X509 Certificates -> Issued Certificates** in the navigation pane on the left of the page. Click on the certificate you'd like to revoke and select the "Revocation" tab in the modal. Choose an option from the "Revocation Reason" dropdown and click "Revoke".

6.2.2 Unrevoke a Certificate

A certificate can only be unrevoked if it was revoked using the certificate hold revocation reason. To unrevoke a certificate, navigate to **X509 Certificates -> Issued Certificates** in the navigation pane on the left of the page. Click on the certificate you'd like to unrevoke and select the "Revocation" tab in the modal. If the certificate was revoked using the "certificateHold" revocation reason, an "Unrevoke" button will be available to undo the revocation.

Note: Any revocation reason other than "certificateHold" will result in a permanent revocation that cannot be undone.

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7. Certificate Issuance Workflows

7.1 Overview

PKIaaS.io supports multiple workflows for issuing certificates. See the links below for more information on each workflow:

7.1.1 Manual Workflows

- Sign a CSR from the Admin UI

7.1.2 Automated Workflows

- Issue Certificates via ACME
- Issue Certificates via SCEP

Note: All expired certificates are automatically removed from the system 30 days after expiration. This ensures the certificate management system remains performant and only retains valid, relevant certificates.

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7.2 Sign a CSR from the Admin UI

To sign a CSR from the admin UI, log in to PKIaaS.io, and navigate to **Certificate Templates -> Manage Templates**. Click on the template you'd like to use, and select "Submit CSR" from the menu. Paste the contents of the CSR into the input field and click "Submit". Before the CSR is signed, it must be reviewed and approved by navigating to **X509 Certificates -> Pending Requests**. See Pending Requests for more information on reviewing and approving pending requests.

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8. ACME

8.1 Overview

The ACME (Automatic Certificate Management Environment) protocol is a standard for automating the process of obtaining, renewing, and managing SSL/TLS certificates. It was developed by the Internet Security Research Group (ISRG), the organization behind Let's Encrypt, and is specified in RFC 8555. PKIaaS.io supports version 2 of the ACME protocol with HTTP-01 and DNS-01 challenges.

Note: PKIaaS.io requires use of an EAB (External Account Binding) to register an ACME account. Because of this requirement, only ACME clients that support EAB will work with the PKIaaS.io ACME service.

A few steps must be completed before requesting a certificate with an ACME client:

1. Create an ACME Policy
2. Generate EAB Credentials
3. Register an ACME Account

The ACME directory URL for all PKIaaS.io ACME requests is https://acme-v02-api.pkiaas.io/directory

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8.2 Create an ACME Policy

An ACME policy associates ACME certificate requests with a specified certificate template and allows a comma-separated list of "pre-approved domains" to be supplied. Pre-approved domains will be automatically authorized and will not be required to do any ACME HTTP or DNS challenges. Wildcard domains are supported to pre-approve an entire root domain and all subdomains.

To create an ACME policy, log in to PKIaaS.io and navigate to **ACME -> Policies**. Click "Create", enter a unique name for the new ACME policy, select which template should be used when signing certificates requested using this policy, optionally enter any pre-approved domains, and click "Create Policy."

Next, see Generate EAB Credentials.

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8.3 Generate EAB Credentials

An ACME policy must be created before EAB credentials can be generated. See Create an ACME Policy if this step hasn't been completed yet.

The use of EAB (external account binding) allows PKIaaS.io to associate ACME accounts with a specific user's PKIaaS.io account, CA, and ACME policy. To generate an EAB for registering a new ACME account, log in to PKIaaS.io and navigate to **ACME -> Policies**. Click the gears in the "External Account Bindings" column of one of the ACME policies to navigate to the EAB management page for the policy and click "Create". Enter a unique name for the EAB. Enter the desired value for "Allowed Use Count". This value represents the number of times this EAB can be used to register unique ACME accounts. Lastly, enter the expiration date after which ACME new account registrations will no longer be authorized with this EAB and click "Create EAB".

When the new EAB is generated, a modal will appear providing the ACME directory URL, the EAB KID and HMAC Key, and some example commands to use the values provided to register a new ACME account using the certbot ACME client. These values can be used to register an ACME account with any ACME client that supports external account bindings. The KID and HMAC Key will only be displayed once, so be sure to copy them before closing the modal.

Note: After an ACME account is registered with an EAB, it is permanently associated with the policy the EAB was generated for. It is safe to delete an EAB after an ACME account has been registered with it.

Next, see Register an ACME Account.

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8.4 Register an ACME Account

An ACME EAB credential must be generated before an ACME account can be registered with PKIaaS.io. See Generate EAB Credentials if this step hasn't been completed yet.

In order to use the ACME protocol to request a certificate from PKIaaS.io, an ACME account must first be registered using an EAB. See the links below for more information about getting started with various ACME clients.

8.4.1 ACME Clients

- certbot
- Certify the Web
- Traefik

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8.5 ACME Clients

8.5.1 Certbot

Be sure to have the EAB KID and HMAC Key ready for the steps below. See Generate EAB Credentials for more information.

All values that appear in ALL CAPS in the examples below must be customized with values specific to you.

Register an Account Only

```
certbot register --server https://acme-v02-api.pkiaas.io/directory --email YOUREMAIL@YOURDOMAIN.COM --agree-tos --eab-kid
YOUREABKID --eab-hmac-key YOUREABHMACKEY
```

Register an Account and Order a Certificate for Apache

```
certbot --apache --server https://acme-v02-api.pkiaas.io/directory --email YOUREMAIL@YOURDOMAIN.COM --agree-tos -d MYDOMAIN.COM
-d WWW.MYDOMAIN.COM --eab-kid YOUREABKID --eab-hmac-key YOUREABHMACKEY
```

See the EFF certbot website for more information about certbot.

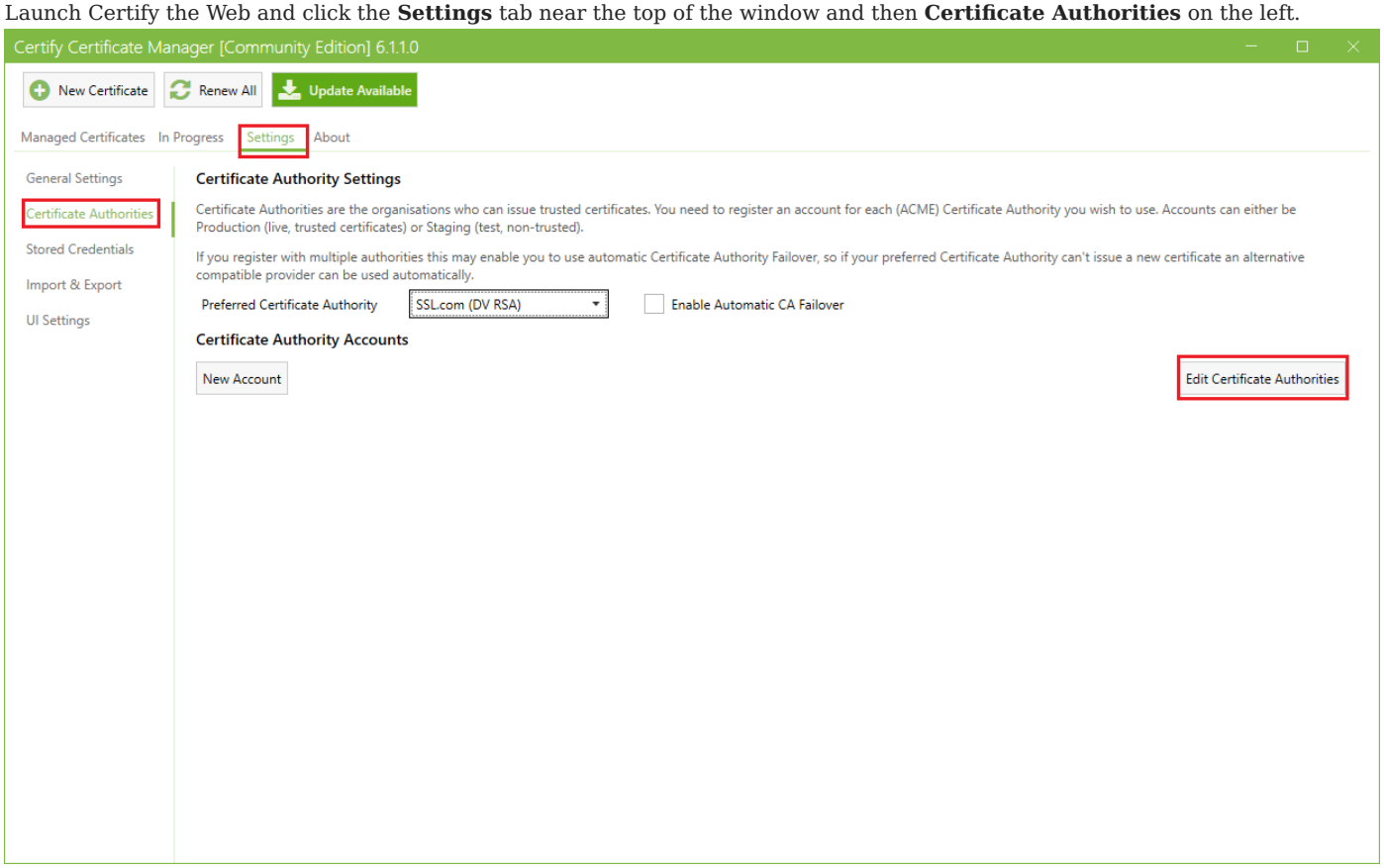
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8.5.2 Certify the Web

Be sure to have the EAB KID and HMAC Key ready for the steps below. See Generate EAB Credentials for more information.

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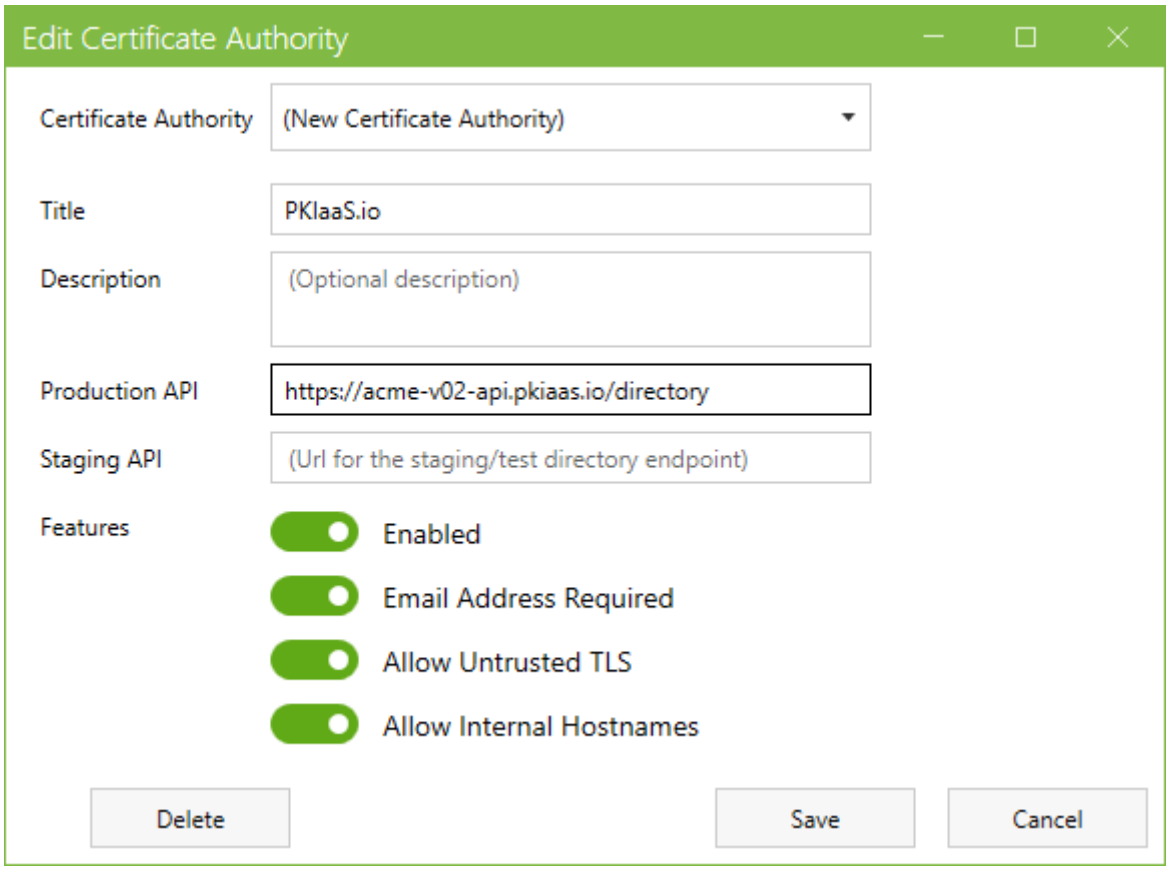
1. Download Certify the Web from https://certifytheweb.com/ and install it.
2. Launch Certify the Web and click the **Settings** tab near the top of the window and then **Certificate Authorities** on the left.



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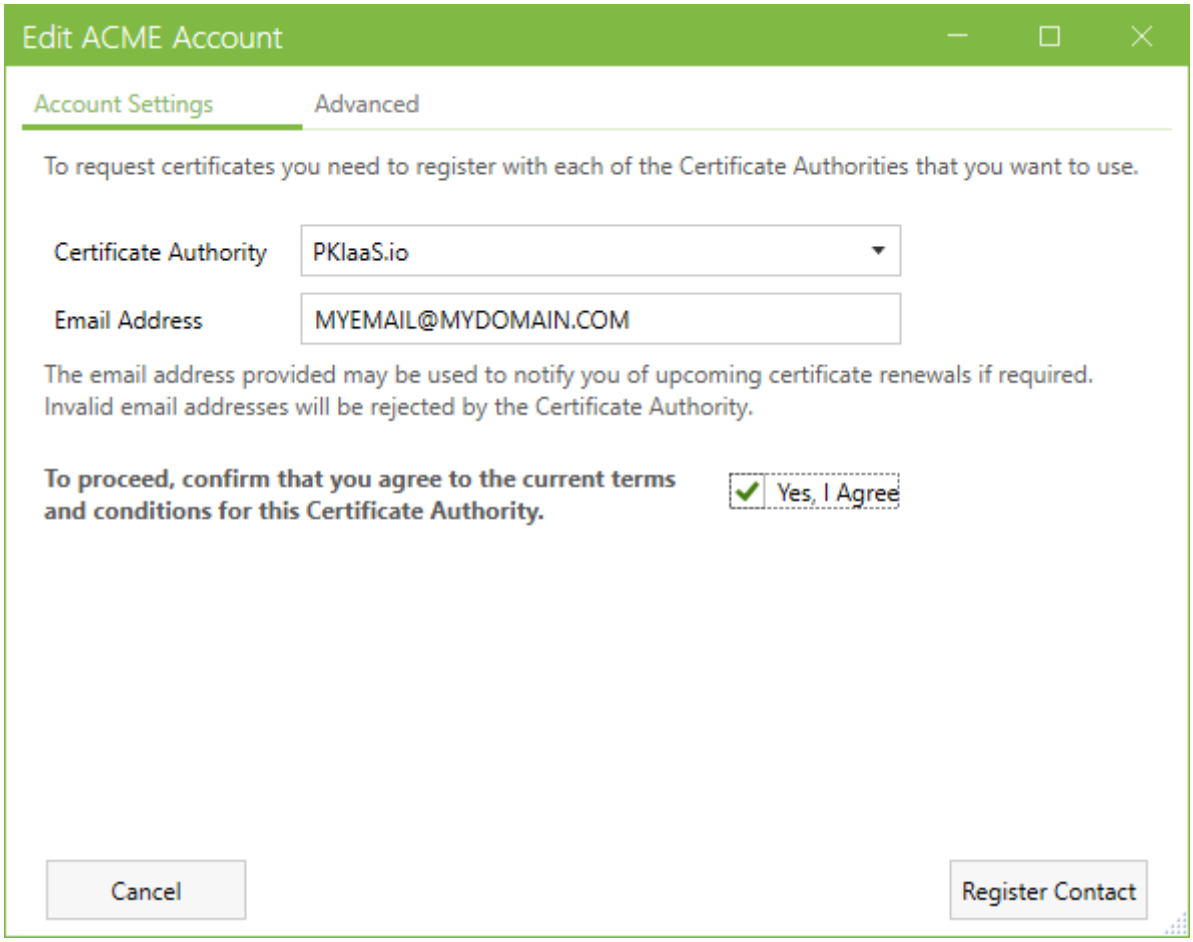
3. Click **Edit Certificate Authorities**.

- In the **Certificate Authority** field, leave "(New Certificate Authority)" selected.
- In the **Title** field, enter "PKIaaS.io" or something else to uniquely identify the new CA.
- In the **Production API** field, enter the PKIaaS.io ACME directory URL "https://acme-v02-api.pkiaas.io/directory".
- Optionally enter a **Description**. The **Staging API** field may be left blank.
- Enable all of the options under **Features**.
- Click **Save**.



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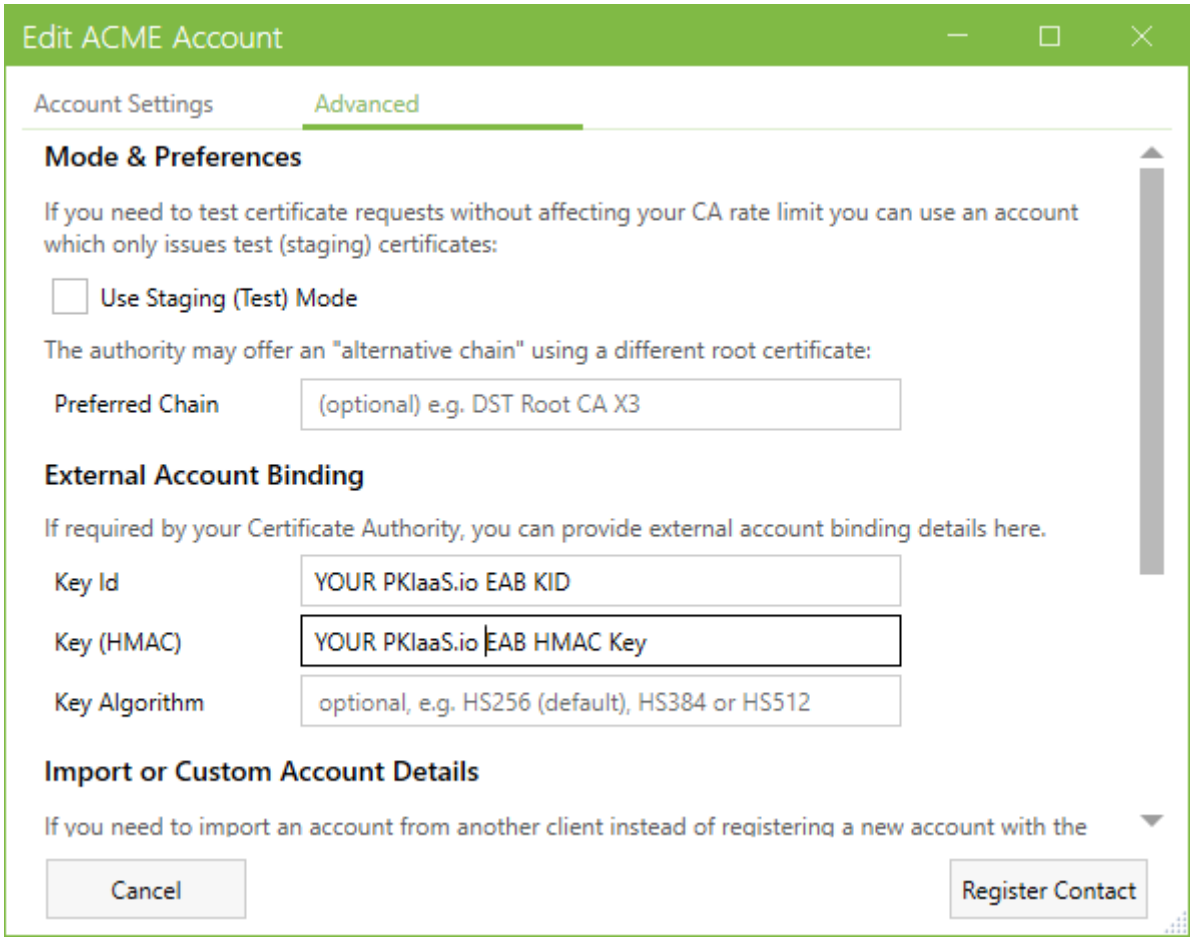
4. Click the **New Account** button on the far left of the Edit Certificate Authorities button clicked in step 3.
   - From the **Certificate Authority** dropdown, select the CA title that was entered in step 3.
   - Enter the **Email Address** to be associated with the new ACME account.
   - Check **Yes, I Agree** to agree to the terms and conditions for the CA.



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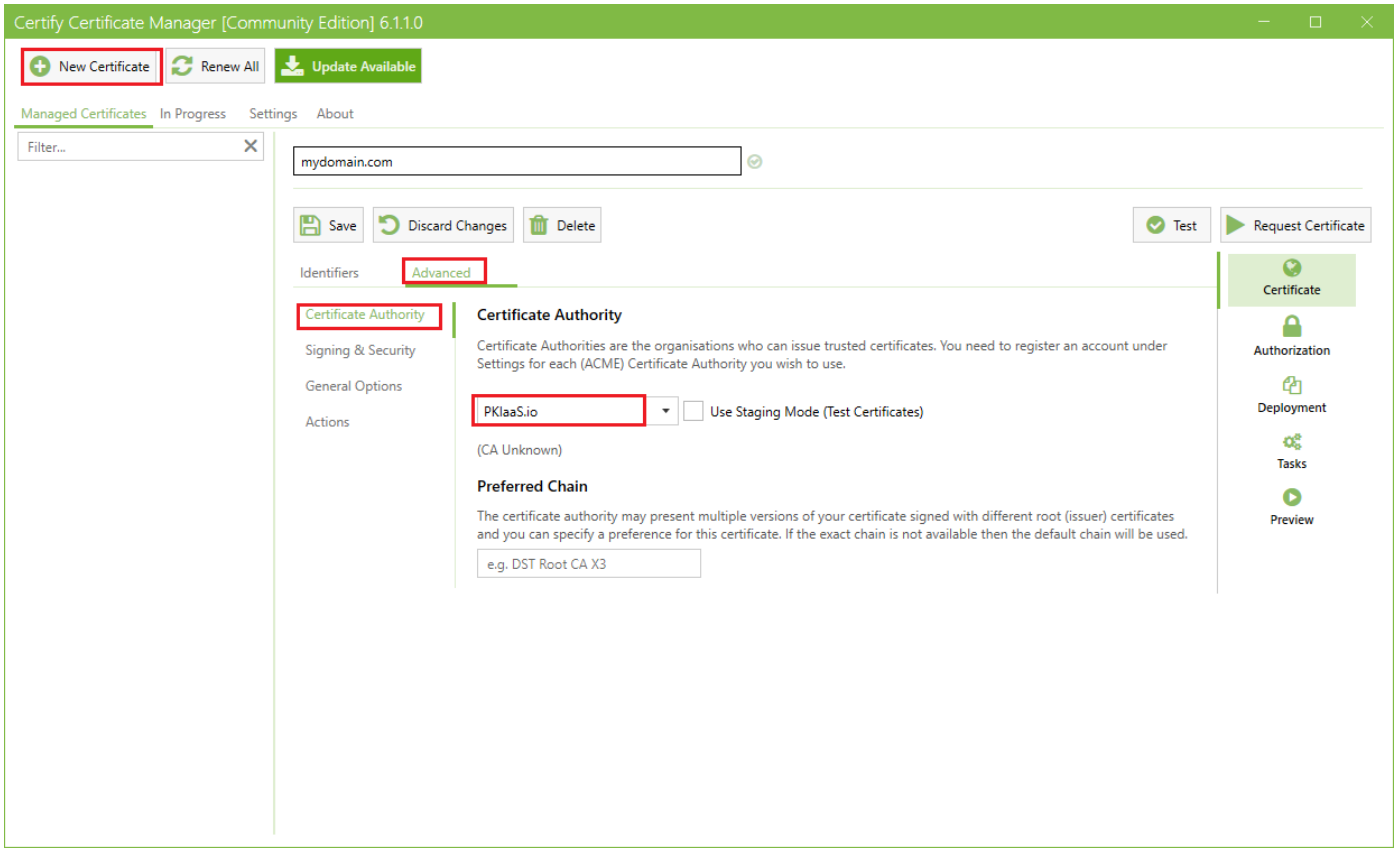
5. Click **Advanced** to enter the EAB credentials.

- Enter the **Key Id** and **Key (HMAC)** using the "EAB KID" and "EAB HMAC Key" provided when generating the EAB credentials.
- Click **Register Contact**.



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6. The ACME account is now registered and ready to request certificates from PKIaaS.io. Be sure to select the new CA created in step 3 when creating a new certificate request.



See Certify the Web's Certificate Manager Documentation for more information about Certify The Web.

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8.5.3 Traefik

Be sure to have the EAB KID and HMAC Key ready for the steps below. See Generate EAB Credentials for more information.

All values that appear in ALL CAPS in the examples below must be customized with values specific to you.

1. Create a docker-compose.yml on your server with the following content:

```
version: "3.3"

services:

  traefik:
    image: "traefik:v3.2"
    container_name: "traefik"
    command:
      #- "--log.level=DEBUG"
      - "--api.insecure=true"
      - "--providers.docker=true"
      - "--providers.docker.exposedbydefault=false"
      - "--entryPoints.web.address=:80"
      - "--entryPoints.websecure.address=:443"
      - "--certificatesresolvers.myresolver.acme.httpchallenge=true"
      - "--certificatesresolvers.myresolver.acme.httpchallenge.entrypoint=web"
      - "--certificatesresolvers.myresolver.acme.caserver=https://acme-v02-api.pkiaas.io/directory"
      - "--certificatesresolvers.myresolver.acme.eab.kid=YOUREABKID"
      - "--certificatesresolvers.myresolver.acme.eab.hmacencoded=YOUREABHMACKEY"
      - "--certificatesresolvers.myresolver.acme.email=YOUREMAIL@YOURDOMAIN.COM"
      - "--certificatesresolvers.myresolver.acme.storage=/letsencrypt/acme.json"
    ports:
      - "80:80"
      - "443:443"
      - "8080:8080"
    volumes:
      - "./letsencrypt:/letsencrypt"
      - "/var/run/docker.sock:/var/run/docker.sock:ro"

  whoami:
    image: "traefik/whoami"
    container_name: "simple-service"
    labels:
      - "traefik.enable=true"
      - "traefik.http.routers.whoami.rule=Host(`whoami.example.com`)"
      - "traefik.http.routers.whoami.entrypoints=websecure"
      - "traefik.http.routers.whoami.tls.certresolver=myresolver"
```

2. Create a "letsencrypt" directory in the same directory you created the docker-compose.yml file.
3. Replace the values for **certificatesresolvers.myresolver.acme.eab.kid**, **certificatesresolvers.myresolver.acme.eab.hmacencoded**, and **certificatesresolvers.myresolver.acme.email** with your own correct values.
4. Replace whoami.example.com with your own domain within the traefik.http.routers.whoami.rule label of the whoami service.
5. Optionally uncomment the following line if you want to test/debug:

```
#- "--log.level=DEBUG"
```

6. Run `docker-compose up -d` within the folder where you created the previous file.
7. Wait a few moments for the certificate request to complete, and visit https://your_own_domain to confirm everything worked as expected.

See the Traefik website for more information about Traefik.

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9. SCEP

9.1 Overview

SCEP (Simple Certificate Enrollment Protocol) is a long-established, open-source protocol that automates the request and issuance of digital certificates from a certificate authority over HTTP. SCEP is often used by mobile device management (MDM) solutions to issue certificates to managed devices. It is also often used by network devices, such as routers and switches, to obtain certificates for secure communication.

9.1.1 Enabling SCEP

PKIaaS.io supports SCEP for certificate enrollment, but it is not enabled by default. SCEP can be enabled and disabled on a per-template basis. To enable SCEP for a template, log in to PKIaaS.io and navigate to **Certificate Templates -> Manage Templates**. Click on the template you wish to enable SCEP for, then click "Edit Template" and select the "SCEP" tab. From here, you can enable SCEP and configure the SCEP settings for the template.

9.1.2 SCEP URL

The SCEP URL is unique to each template. To find the SCEP URL for a template, navigate to **Certificate Templates -> Manage Templates**, click on the template you wish to use and select "Show Template Service URLs" from the menu. This is the URL that devices will use to communicate with PKIaaS.io to request a certificate from the template.

The SCEP URL is available over both HTTPS and HTTP. Use HTTPS whenever the device supports it, since it also protects the device's initial download of the CA certificate. The HTTP URL is provided for devices that don't support HTTPS.

9.1.3 SCEP Validation

When a device requests a certificate via SCEP, PKIaaS.io will validate the request before issuing the certificate. Multiple validation methods are supported. See the links below for more information on each.

1. Static Passphrase
2. Microsoft NDES Compatibility
3. Intune Integration

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9.2 Intune Integration

PKIaaS.io natively supports SCEP integration with Microsoft Intune, allowing certificates to be validated and issued to devices managed by Intune. The integration can be connected automatically with one click, or configured with an app registration you create in Microsoft Entra ID.

9.2.1 Enabling Intune Integration

To enable Intune integration, log in to PKIaaS.io, and navigate to **Certificate Templates -> Manage Templates**, click on the template you wish to enable Microsoft Intune integration for, then click "Edit Template" and select the "SCEP" tab. Check the "Enable Microsoft Intune/Endpoint Manager Integration" checkbox, then connect automatically or use your own app registration.

Connect Automatically (Recommended)

Click "Connect to Intune". You will be redirected to Microsoft to sign in as a Global Administrator or Privileged Role Administrator for your tenant and approve the PKIaaS.io Intune Connector. The connector requests **SCEP challenge validation** (Intune), **Read all applications** (Microsoft Graph, used to locate your tenant's Intune SCEP validation service), and **Sign in and read user profile** (used to confirm which tenant approved the connector). After approving, you will be signed in to your tenant to confirm the connection, and then returned to the template, which will show that it is connected. No app registration or client secret is needed.

Note: Save any other changes to the template before clicking "Connect to Intune". If you are signed in to more than one Microsoft account, you will be asked to pick your account twice. Microsoft may take a few minutes to activate a new connection. If another template on the same certificate authority is already connected, click "Use tenant" to connect without approving again.

To disconnect, uncheck "Enable Microsoft Intune/Endpoint Manager Integration" and click "Save Template". Deleting the PKIaaS.io Intune Connector from **Enterprise applications** in Microsoft Entra ID disconnects every template using your tenant.

Use Your Own App Registration

Click "Use my own app registration instead", then create an app registration by following Microsoft's instructions to authorize communication between CA and Intune. Enter the following information:

- **Tenant ID:** your tenant ID or primary domain (e.g. contoso.onmicrosoft.com)
- **Application (client) ID:** found on the app registration's Overview page
- **Client Secret:** the client secret value created under **Certificates & secrets**

Finally, click "Save Template" to enable the integration.

Note: Client secrets expire. When the secret expires, Intune validation will fail until a new client secret is saved on the template, or the template is connected automatically.

9.2.2 Create and assign SCEP certificate profiles in Intune

After the integration is enabled, the remainder of the configuration is done in Intune. Follow the steps at https://learn.microsoft.com/en-us/intune/device-configuration/certificates/scep-profiles to create and assign SCEP certificate profiles in Intune. Be sure to have the template SCEP URL handy while configuring the SCEP certificate profile. See SCEP Overview for help finding the URL for the template.

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9.3 Microsoft NDES Compatibility

PKIaaS.io offers Microsoft NDES compatibility to support third-party applications and processes built to work with Microsoft NDES. The feature can be enabled on a per-template basis and can be used to generate one-time challenge passwords for certificate requests from the template. The challenge passwords expire after 60 minutes.

9.3.1 Enable Microsoft NDES Compatibility

To enable Microsoft NDES compatibility for a template, navigate to **Certificate Templates -> Manage Templates**, click on the template you wish to enable Microsoft NDES compatibility for, then click "Edit Template" and select the "SCEP" tab. Check the "Enable Microsoft SCEP Compatibility" checkbox and save the template.

9.3.2 Microsoft NDES Compatibility URL

After enabling Microsoft NDES compatibility, the URL to access the Microsoft compatible UI can be found by navigating to **Certificate Templates -> Manage Templates**, clicking on a template, and selecting "SCEP Admin (Microsoft Compatibility)" from the menu. This URL is unique to each template and can be used to generate one-time passcodes for certificate requests from the template.

9.3.3 Authentication

Access to the Microsoft NDES Compatibility URL is automatically authenticated using the same credentials used to log in to PKIaaS.io if there is an active session. In order to access the URL without an active session—this may be required by third-party applications designed to integrate with Microsoft NDES—basic authentication can be enabled by checking the "Enable Microsoft Basic Authentication" checkbox in the "SCEP" tab of the template configuration. Enter a username and password to use for basic authentication and save the template.

9.3.4 Restrict Access

To restrict access to the Microsoft NDES Compatibility URL to specific IP addresses, see SCEP Security.

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9.4 Apple Configuration Profile

PKIaaS.io supports a feature that allows Apple devices running iOS or macOS to request a certificate via the SCEP protocol without the need for mobile device management. This works by creating an Apple configuration profile and emailing it to an Apple user to download and install on their device. The profile contains the SCEP URL and other configuration settings that the device uses to request a certificate from PKIaaS.io.

9.4.1 Prerequisites

SCEP must first be enabled on the template that will be used to issue certificates to Apple devices. Follow the steps in the SCEP Overview to enable SCEP on the template.

9.4.2 Create an Apple Configuration Profile

To create an Apple configuration profile to request a certificate via SCEP, log in to PKIaaS.io, and navigate to **Certificate Templates -> Manage Templates**, click on the template you wish to issue the certificate from, and click "Create Pre-Approved Request". Fill out the form following the below instructions:

1. Enter the desired subject alternative names, common names, and other subject attributes
2. Choose the "Apple Device" option
3. Enter the email address of the Apple user that will receive the profile
4. Click "Submit"

9.4.3 Download and Install the Apple Configuration Profile

After submitting the form, an email will be sent to the Apple user with a link to download the configuration profile. The user can then download and install the profile on their device and the device will automatically request a certificate from PKIaaS.io using the SCEP protocol. See the documentation from Apple in the links below for more information on how to install the profile on an Apple device after it has been downloaded.

- Install a configuration profile on iOS, iPadOS, and visionOS devices
- Install a configuration profile on macOS devices

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9.5 Static Passphrase

PKIaaS.io supports the use of a static passphrase as a validation method for SCEP certificate requests; however, using a static passphrase is generally considered a security risk due to its potential for unauthorized access if compromised. For this reason, we recommend using a more secure validation method, such as Microsoft NDES Compatibility or Intune Integration, whenever possible.

9.5.1 Enable Static Passphrase

Within the SCEP tab of the template configuration, check the "Enable SCEP Static Passphrase" checkbox. Click the input that says "Click to Generate" to generate a passphrase. This passphrase will not be visible after the template is saved, so be sure to copy it to a secure location before saving.

Note: The passphrase can be updated in the future by clicking the "Click to Generate" input again. This will generate a new passphrase and invalidate the old one. If ever it is clicked by mistake, be sure to click "Cancel" in the template editor to keep the original passphrase.

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9.6 Security

SCEP access can be restricted to specific IP addresses. To restrict access to a specific IP address, navigate to **Certificate Templates -> Manage Templates**, click on the template you wish to restrict access to and select the "SCEP" tab. Check the "Enable IP ACL" checkbox and enter the IP address you wish to allow SCEP requests from in the "Allowed IPs" field. Multiple IP addresses can be entered, separated by commas. Using the "Limit access to" dropdown, you may choose whether to limit access to SCEP requests, the Microsoft compatibility UI, or both.

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10. Windows Integration

10.1 Overview

PKIaaS.io supports Microsoft Certificate Enrollment Policy Web Service and Certificate Enrollment Web Service, also known as CEP and CES, to enable certificate issuance and automatic renewal for Windows devices. This integration allows Windows clients to request certificates from PKIaaS.io seamlessly using the native Windows certificate enrollment features. For more information about CEP and CES, see the Microsoft documentation below:

- Certificate Enrollment Web Services in Active Directory Certificate Services
- MS-WCEP (CEP) Protocol Documentation
- MS-WSTEP (CES) Protocol Documentation
- Client Configuration for Key-Based Renewal

CEP is used to provide certificate enrollment policy information to Windows clients, allowing them to understand which templates are available for certificate requests. CES is used to handle the actual certificate enrollment requests from Windows clients, including the issuance and renewal of certificates. CES is particularly useful for automatic certificate renewal, as it allows Windows clients to renew certificates without user intervention.

10.1.1 Authentication

PKIaaS.io supports Username/Password and X.509 Certificate authentication methods on Windows clients for both Certificate Enrollment Policy Web Service (CEP) and Certificate Enrollment Web Service (CES). X.509 Certificate authentication with the CES service can only be used for certificate renewal. Username and password must be used for new certificate enrollment.

The username and password for CEP and CES can be defined in the CA edit and template edit pages, respectively, in their respective tabs. The username and password can be unique for each service or the option "Use CEP Credentials" can be selected within the template configuration to use the same credentials as CEP. The username and password may also be unique for each template, allowing for granular control over which templates can be used by users.

10.1.2 Enable Windows Integration

CEP can be enabled by logging in to PKIaaS.io and navigating to **Certificate Authorities -> Manage CAs** in the navigation pane on the left side of the page. Click on the CA you want to enable CEP on, and select **Edit CA**. Navigate to the **CEP** tab and select the "Enable Microsoft Certificate Enrollment Policy Web Service (CEP)" checkbox. A mandatory username and password field will appear, allowing you to set the credentials for the CEP service. Click "Save" to apply the changes.

CES can be enabled by navigating to **Certificate Templates -> Manage Templates** in the navigation pane on the left side of the page. Click on the template you want to enable CES on, and select **Edit Template**. Navigate to the **CES** tab and select the "Enable Microsoft Certificate Enrollment Web Service (CES)" checkbox. This option is not available unless CEP has been enabled on the CA the template is associated with. Additional fields will appear, allowing the CES service to be configured. The username and password can be set to the same credentials as CEP by selecting the "Use CEP Credentials" checkbox, or they can be set to unique values. The username and password cannot be left blank if the "Use CEP Credentials" option is not selected. Click "Save" to apply the changes.

CES must be manually enabled on each template that will be used for certificate issuance on Windows clients. All CES-enabled templates will appear on Windows clients when requesting certificates; however, unique usernames and passwords can be set for each template to control access to certificate issuance by each template.

10.1.3 Template Types

CES-enabled "Computer" templates will appear on Windows clients when requesting machine certificates, and CES-enabled "User" templates will appear on Windows clients when requesting user certificates. Be sure to create the appropriate template type for the intended use case. The template type can be set when creating a new template or editing an existing template by

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navigating to the **CES** tab within the template edit page. The "Template Type" dropdown allows you to select either "Computer" or "User" as the template type.

10.1.4 Cryptography Options

The cryptography options for CES-enabled templates can be configured by navigating to the **CES** tab within the template edit page. The "Cryptography" dropdown allows you to select the cryptographic algorithms that will be available for use when requesting certificates with the template. The "Minimum Key Size" dropdown allows you to set the minimum key size that will be enforced when requesting certificates. The minimum key size dropdown may not be present, depending on the selected cryptographic algorithms. A list of Cryptographic Storage Providers can be selected to limit the available CSPs for use when requesting certificates with the template. Only the selected CSPs will be available for use when requesting certificates with the template. If none are selected, all CSPs will be available.

10.1.5 Additional Options

The "Allow the private key to be exported" checkbox allows the private key to be exported from the certificate. This is useful for scenarios where the private key needs to be backed up or used on multiple devices.

The "Enable strong private key protection" checkbox enables strong private key protection, which requires the user to enter a password to access the private key. This is useful for scenarios where the private key needs to be protected from unauthorized access.

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10.2 Add Enrollment Policy

Before adding an enrollment policy and configuring Windows to request certificates from PKIaaS.io, CEP and CES must be enabled on your CA and templates. See Enabling Windows Integration if this step hasn't been completed yet.

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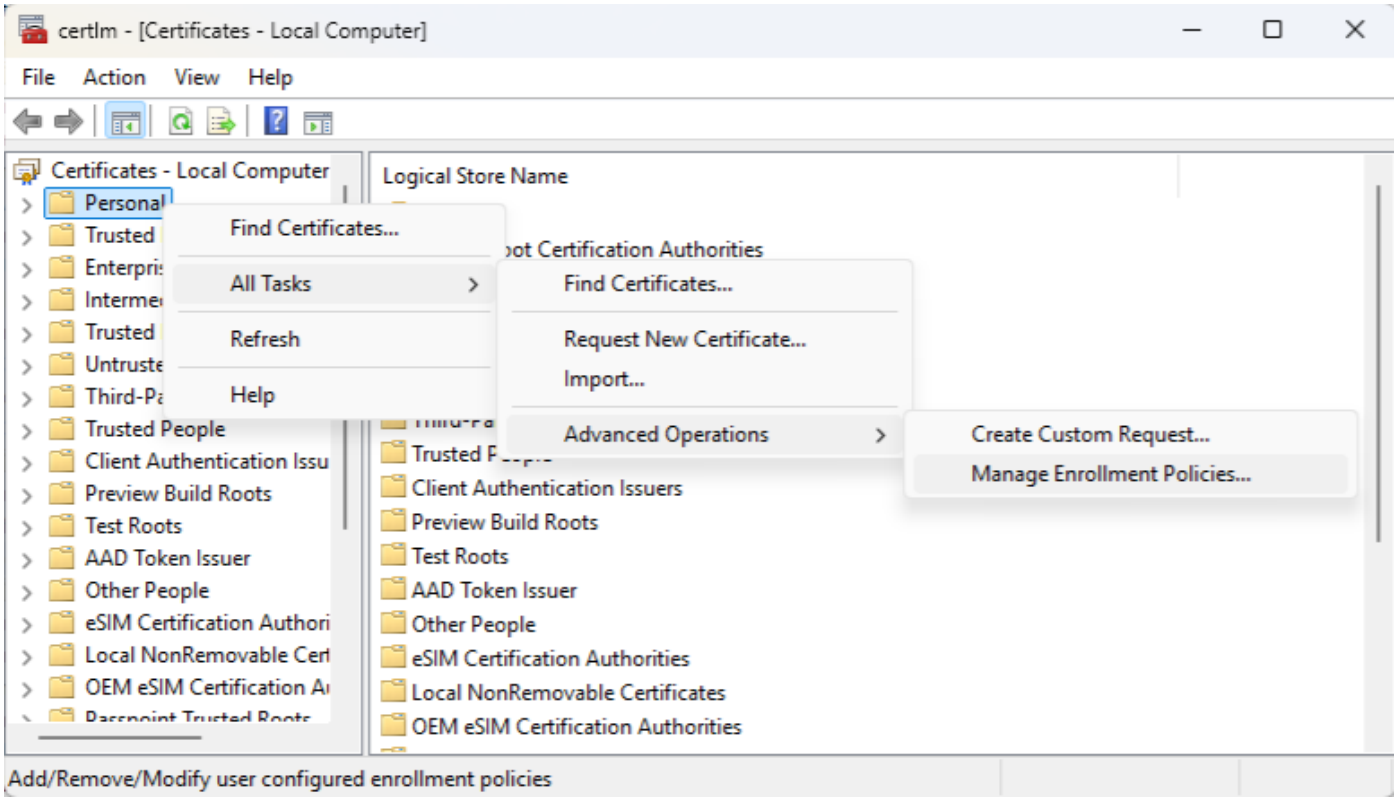
10.2.1 Client Configuration

To configure Windows clients for certificate issuance and renewal with PKIaaS.io, follow these steps:

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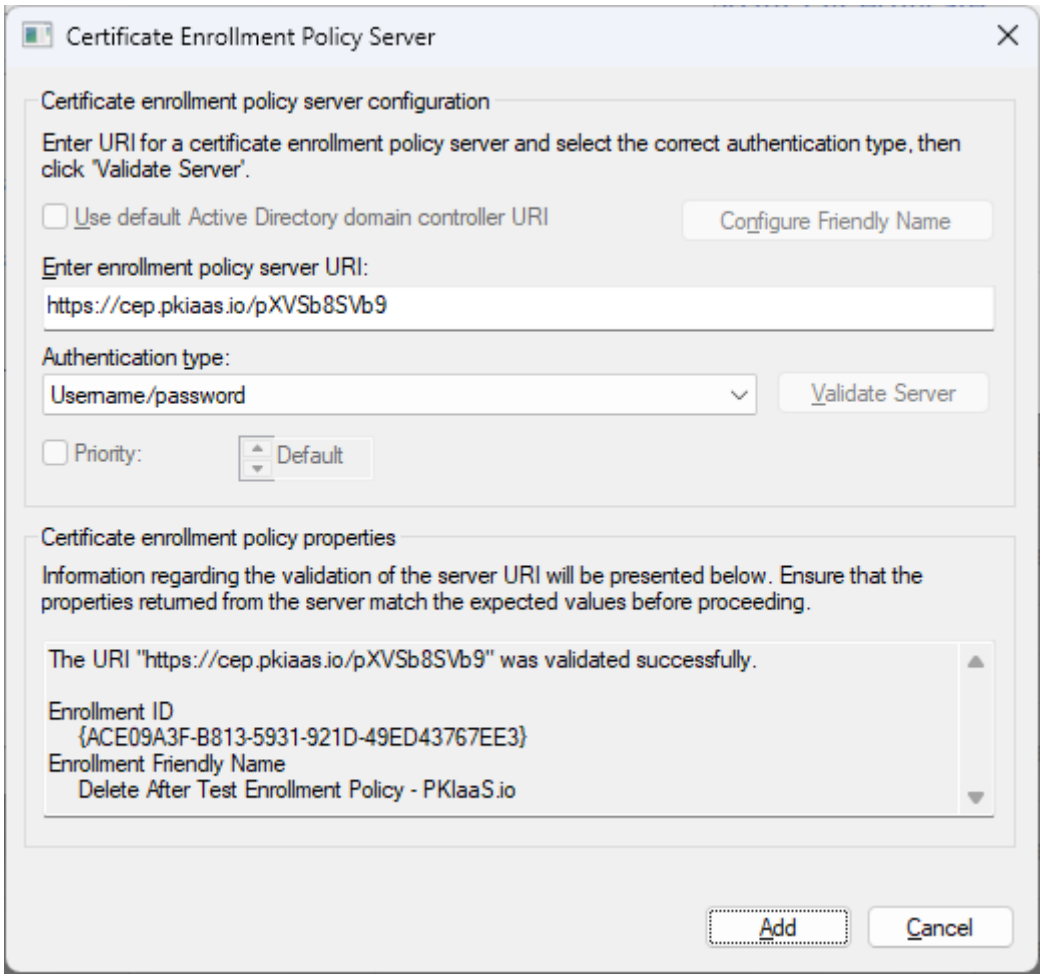
1. **Install the Root CA Certificate**: Download the root CA certificate from PKIaaS.io and install it on the Windows client. This is necessary to trust the certificates issued by PKIaaS.io. The root CA certificate can be downloaded by navigating to **Certificate Authorities -> Manage CAs** in the navigation pane on the left side of the page. Click on the CA you want to download, and select **Get CA Certificate(s)**. A download option is available beneath the "Certificate" tab. Import the certificate into the "Trusted Root Certification Authorities" store on the Windows client.
2. **Configure Certificate Enrollment Policy**:

- Use the command certmgr.msc (for user certificates) or certlm.msc (for machine certificates) to open the Certificate Manager. Right-click on "Personal" and select "All Tasks -> Advanced Operations -> Manage Enrollment Policies...". This will open the Certificate Enrollment wizard. Click "Next" to proceed.



- Click "Add" to add a new enrollment policy server. Enter the URL of the PKIaaS.io CEP service. The URL can be found by navigating to **Certificate Authorities -> Manage CAs** in the navigation pane on the left side of the page. Click on a CA, and select **Show CA Service URLs**. Use the "CEP (User/Pass)" or "CEP (Cert Auth)" URL, depending on the authentication method you want to use. Select "Username/password" or "X.509 Certificate" as the authentication type, based on the desired configuration and the URL supplied, and click "Validate Server". You will be prompted to enter the username and password for the CEP service. If you are using X.509 Certificate authentication, you will need to select the certificate to use for authentication. Click "Add" to save the enrollment policy server.

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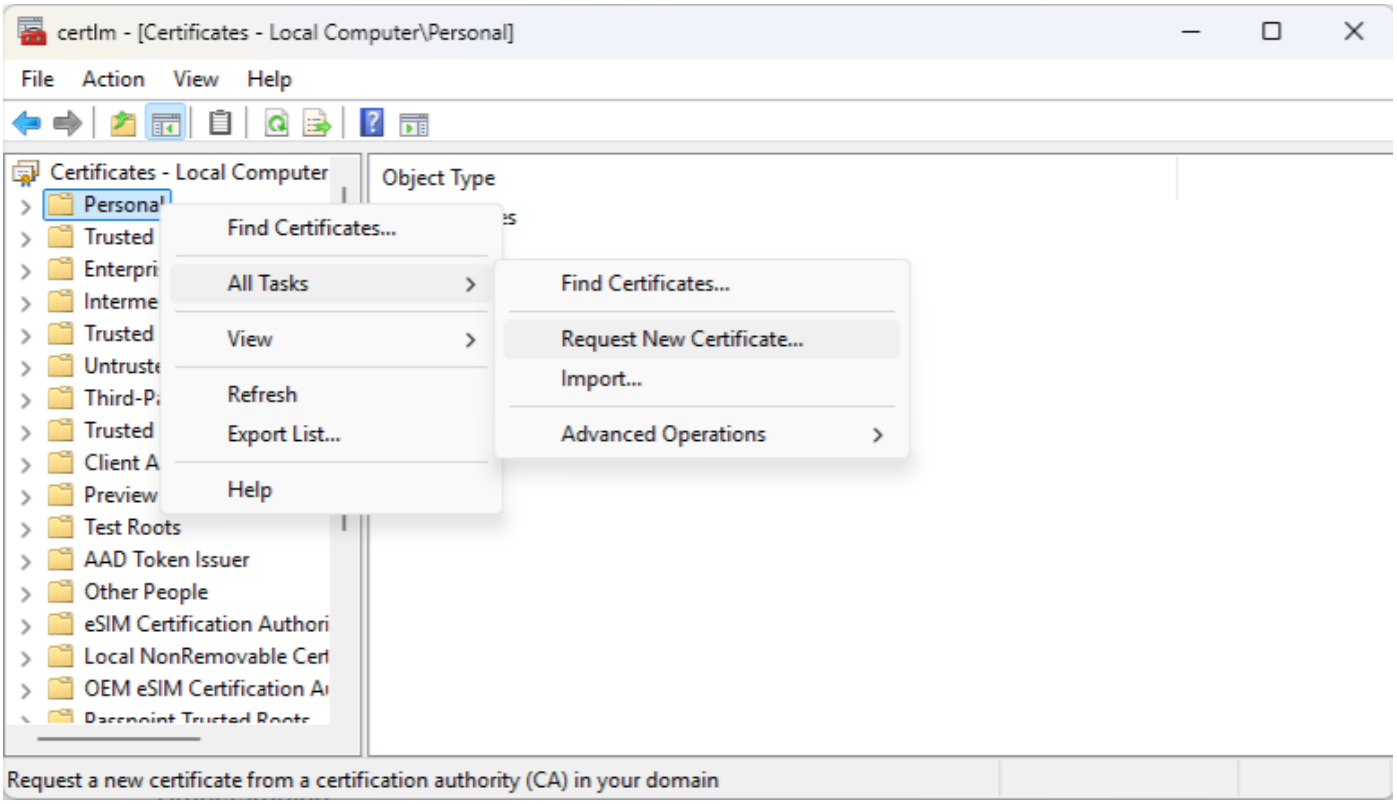
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10.3 Request Certificates

Before requesting certificates using the PKIaaS.io native Windows integration, an enrollment policy must be added to the Windows client. See Add Enrollment Policy if this step hasn't been completed yet.

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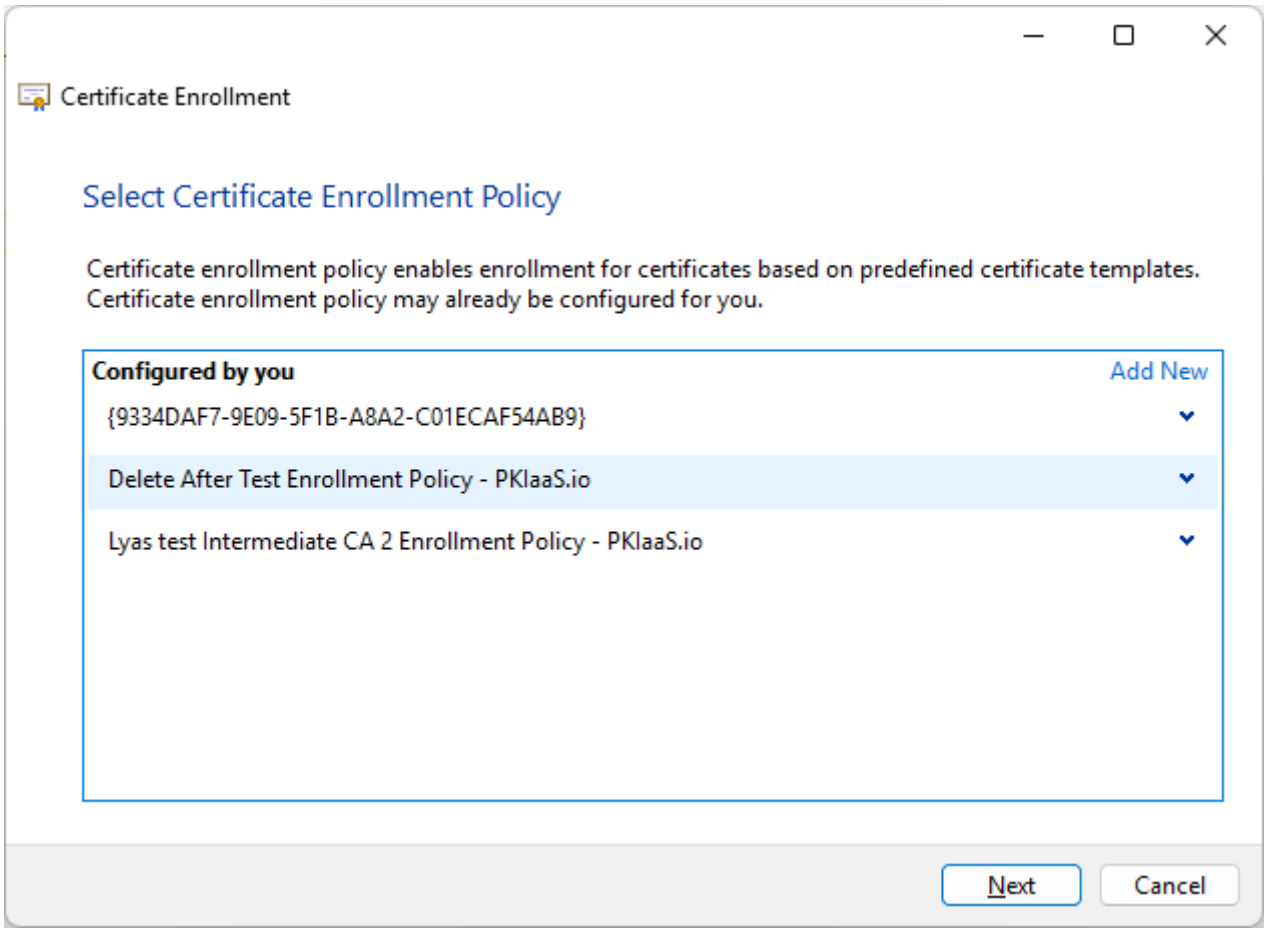
1. Use the command certmgr.msc (for user certificates) or certlm.msc (for machine certificates) to open the Certificate Manager. Right-click on "Personal" and select "All Tasks -> Advanced Operations -> Manage Enrollment Policies...". This will open the Certificate Enrollment wizard.



2. Click "Next" to continue, and select the enrollment policy server that was added previously. You may be prompted to enter the username and password for the CEP service if you are using Username/Password authentication. If you are using X.509 Certificate

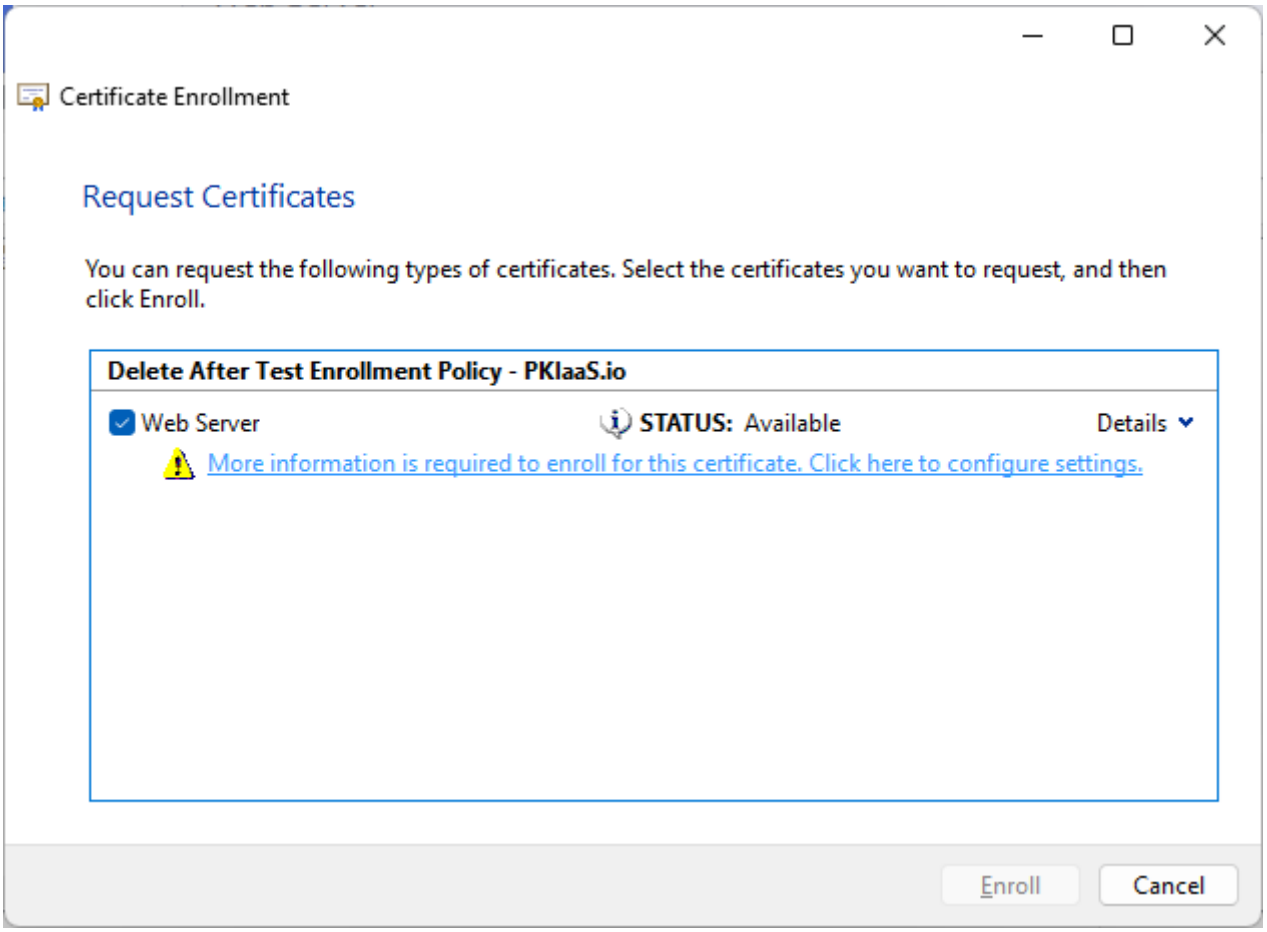
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authentication, select the certificate to use for authentication.



3. A list of CES-enabled certificate templates will be displayed. If the template you want to use is not listed, ensure that the template is CES-enabled and that it is configured with the correct template type (Computer or User) for the certificate you want to request. Select the template you want to use and click "Next".

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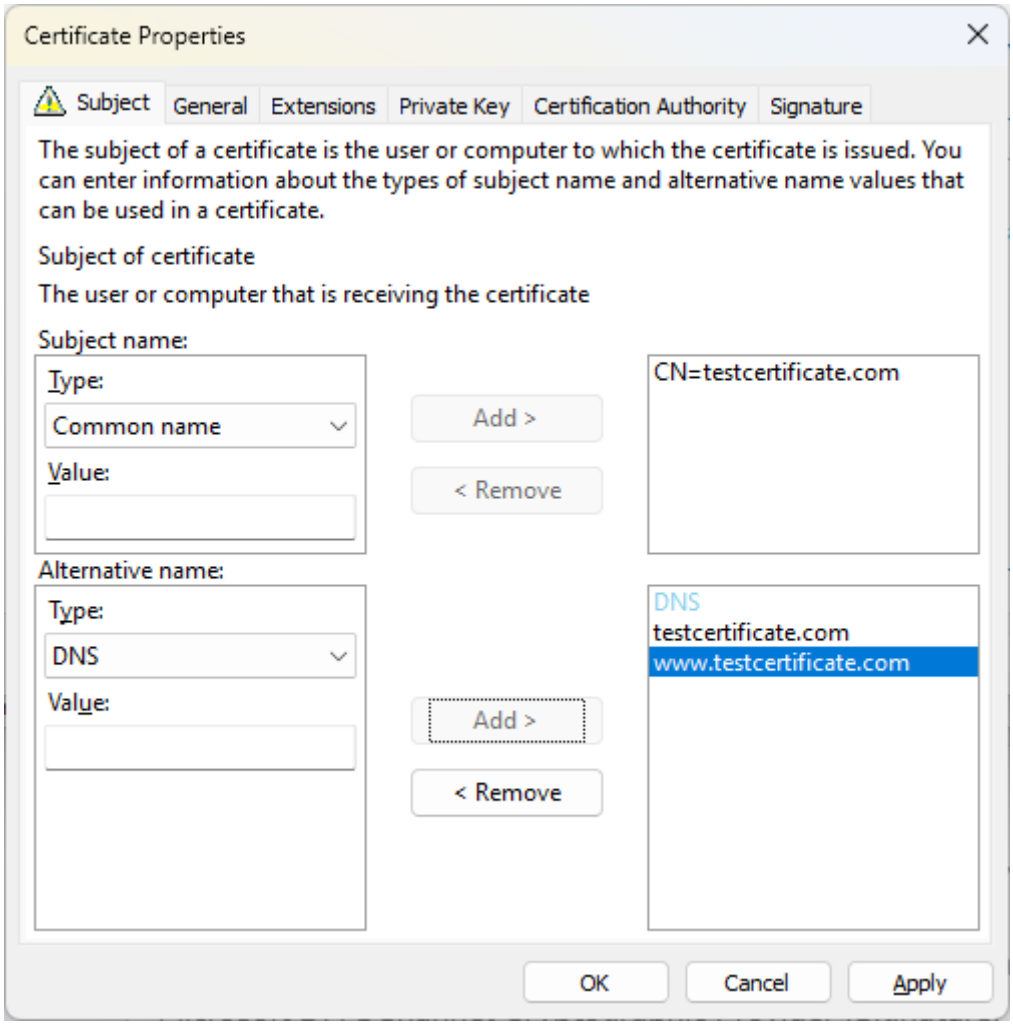


Note: If a template was added or modified recently, it may take up to an hour for the changes to propagate to the Windows client due to caching. For immediate access to the new or modified template, the enrollment policy must be removed and re-added on the Windows client.

4. Click the warning link beneath the template "More information is required to enroll for this certificate. Click here to configure settings." This will open the certificate properties dialog where you can configure the certificate request settings including subject attributes, subject alternative names (SANs), and cryptographic options. The options available will depend on the template

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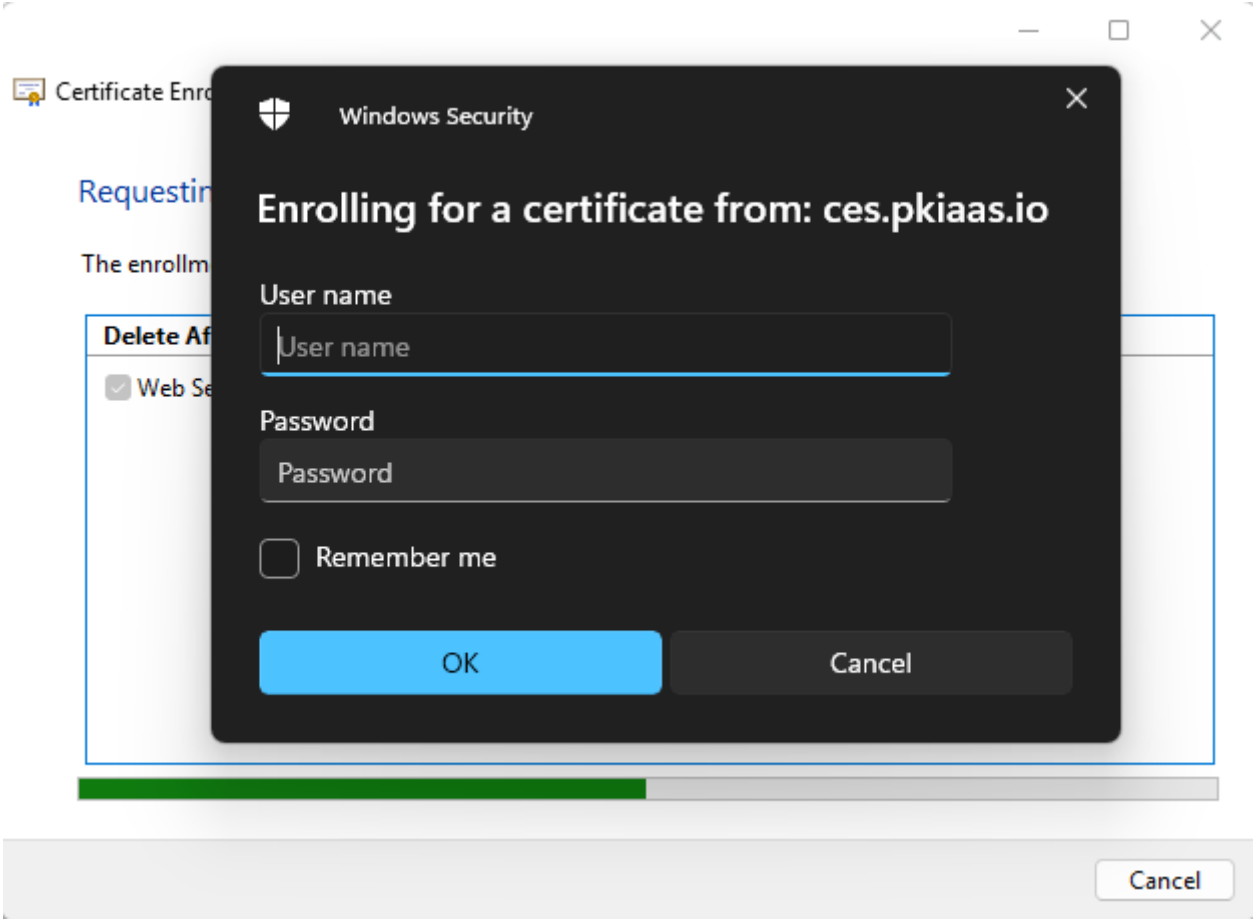
configuration. Once you have configured the settings, click "OK" to close the dialog.



5. Finally, click "Enroll" to request the certificate. The certificate will be requested from the PKIaaS.io CES service using the selected template and the configured settings. Unless it is cached from a previous request, you will be prompted to enter the username and password associated with the template for the CES service. If the request is successful, the certificate will be issued and added to

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the "Personal" store in the Certificate Manager.



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11. OCSP

11.1 Overview

OCSP (Online Certificate Status Protocol) is a network protocol used in Public Key Infrastructure (PKI) to check the revocation status of digital certificates. It allows clients, like web browsers, to verify whether a certificate is still valid or has been revoked without requiring a full download of a Certificate Revocation List (CRL).

11.2 OCSP Signing Certificate

By default, PKIaaS.io generates a certificate with the "OCSP Signing" extended key usage (EKU) attribute for each CA when the CA is created. The certificate will appear in **X509 Certificates -> Issued Certificates** with the common name "PKIaaS.io OCSP Responder". This certificate is used to sign OCSP responses for the CA. When the certificate expires or is manually revoked, a new OCSP signing certificate will be generated on demand during the next OCSP request.

11.3 OCSP Responder URL

To find the OCSP responder URL along with many other service URLs, log in to PKIaaS.io, and navigate to **Certificate Authorities -> Manage CAs**. Click on the desired CA, and select "Show CA Service URLs". A valid OCSP request must be sent to the OCSP responder URL to receive a signed OCSP response. Tools like the OCSP Checker can be used to return the OCSP response for a certificate.

11.4 OCSP in the AIA Extension

By default, the OCSP URL is also encoded in issued certificates in the Authority Information Access (AIA) extension. This can be changed within a certificate template by navigating to **Certificate Templates -> Manage Templates**, clicking on a template, and selecting "Edit Template". The OCSP URL can be removed from certificates issued using the template by unchecking "Include OCSP Extension in Certificate" in the **CRL / OCSP / AIA / CPS** section of the template settings.

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12. CRL

12.1 Overview

A CRL (Certificate Revocation List) is a list published by a Certificate Authority (CA) that contains the serial numbers of digital certificates that have been revoked before their scheduled expiration. This list is part of Public Key Infrastructure (PKI) and serves as a mechanism to help entities verify whether a certificate is still valid.

12.2 CRL Generation

PKIaaS.io automatically generates CRLs on demand when an HTTP request is received to download the CRL. CRLs are cached and valid for a period of 30 days from the time of generation. When the CRL expires or another certificate is revoked, the cache is deleted, and a new CRL is generated on demand during the next HTTP request to download the CRL.

12.3 CRL URL

To find the CRL URL along with many other service URLs, log in to PKIaaS.io, and navigate to **Certificate Authorities -> Manage CAs**. Click on the desired CA, and select "Show CA Service URLs". Click the CRL URL to download the CRL.

12.4 CRL Distribution Point

By default, the CRL URL is also encoded in issued certificates in the CRL Distribution Points (CDP) extension. This can be changed within a certificate template by navigating to **Certificate Templates -> Manage Templates**, clicking on a template, and selecting "Edit Template". The CRL URL can be removed from certificates issued using the template by unchecking "Include CRL Extension in Certificate" in the **CRL / OCSP / AIA / CPS** section of the template settings.

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13. Timestamping

13.1 Overview

Timestamping is the process of adding a trusted and verifiable time and date to a piece of digital data. This establishes proof that the data existed or was signed at a specific moment in time and has not been altered since. It ensures that signed data remains verifiable and trustworthy long after the initial signing event, even after the signing certificate has expired.

13.2 Timestamping Service

PKIaaS.io provides a timestamping service that allows users to request a trusted timestamp for code signing, PDF document signing, etc. Both the RFC 3161 and Authenticode timestamping protocols are supported. The timestamping service URL is unique to each CA and can be found by logging in to PKIaaS.io and navigating to **Certificate Authorities -> Manage CAs**. Click on the desired CA, and select "Show CA Service URLs".

13.3 More Information

See the links below for more information about various tools for timestamping documents and code.

- https://www.digicert.com/kb/code-signing/digicert-certificate-utility-to-sign-code.htm
- https://globaltrust.eu/en/adobe-acrobat-reader-dc-use-certificate-timestamp-server/
- https://learn.microsoft.com/en-us/windows/win32/seccrypto/signtool

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14. Iot-HSM

14.1 Overview

See the installation documentation to get started with IoT-HSM.

IoT-HSM is a lightweight application designed to allow Google Cloud KMS keys and YubiKeys to be used to protect CA private keys on PKIaaS.io. Once deployed and provisioned for a Certificate Authority (CA), it securely forwards all signing requests for the CA from PKIaaS.io to the IoT-HSM, where the configured HSM handles the signing. To ensure robust security, all communications with the IoT-HSM are digitally signed with end-to-end encryption via S/MIME. Advantages of using IoT-HSM include:

- IoT-HSM is an extremely affordable HSM solution for only the cost of the YubiKey (approximately $50) or Google Cloud KMS key (approximately $1 - $3 per month depending on usage)
- Private keys can be imported or generated and never leave the device
- One or more HSMs can be managed by a single IoT-HSM appliance
- Multiple slots on a YubiKey can be provisioned with different private keys, allowing for multiple CAs to be managed by a single YubiKey
- Keys can effortlessly be moved to new hardware or an upgraded install of the IoT-HSM appliance
- Multiple IoT-HSM appliances can be used to provide redundancy and load balancing

Note: IoT-HSM also offers the ability to create a virtual HSM using SoftHSM2, which can be used for testing and development purposes but is not recommended for production use. To create a SoftHSM2 virtual HSM, click "Create SoftHSM2" on the IoT-HSM dashboard.

Next, see Create an HSM CA to create a CA for use with IoT-HSM.

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14.2 Installation

14.2.1 Virtual or Physical Machine

When installing IoT-HSM, it is recommended to deploy a Long-Term Support (LTS) version of Ubuntu Server. IoT-HSM has been tested on Ubuntu 24.04 and 25.10. Once Ubuntu is installed and updated, log in to a shell, and execute the following commands as the root user. You can become root by entering the command `sudo su -`.

```
apt update
apt -y install curl
curl -sL https://raw.githubusercontent.com/lspiehler/iot-hsm/master/scripts/setup_ubuntu.sh | bash -
```

After the script completes, it will prompt to set a new password twice. This password will be used for the admin account on the IoT-HSM web interface. Once configured, open a browser and navigate to the IP address of the Ubuntu endpoint, prefixed with `https://`. For example: `https://192.168.1.5`.

See the Create an HSM CA documentation to create a certificate authority for use with IoT-HSM.

14.2.2 Container Deployment

Below is an example command that can be used to deploy IoT-HSM as a container.

```
docker run -it -d --restart=always --name iot-hsm -e LISTENIP=0.0.0.0 -p 3000:3000 -v iot-hsm:/var/lib/softhsm/tokens -v /var/node/iot-hsm/state:/var/node/
iot-hsm/state ghcr.io/lspiehler/iot-hsm:latest
```

IoT-HSM was designed to support all container platforms like those provided by GCP, Azure, and AWS. Advanced knowledge of those platforms may be required for successful deployment. Be sure to mount the appropriate volumes for persistent storage.

No authentication or TLS support is provided by the IoT-HSM containerized deployment. Additional containers (e.g., nginx or Traefik reverse proxies deployed as a sidecar) or cloud resources (load balancers, application gateways, etc.) are required to provide these features if desired.

Note: YubiKeys are not supported on IoT-HSM container deployments at this time.

See the Create an HSM CA documentation to create a certificate authority for use with IoT-HSM.

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14.3 Create an HSM CA

If you haven't yet deployed IoT-HSM, see the installation documentation to get started.

Create a CA with an HSM Key on PKIaaS.io

In order for a certificate authority on PKIaaS.io to delegate signing operations to an HSM managed by IoT-HSM, it must first be provisioned to do so. The first step in this process is to select the "HSM" option as the "Key Type" when creating a new certificate authority on the **Certificate Authorities -> Create new CA** page.

Retrieve HSM Provisioning Certificate from PKIaaS.io

After clicking "Create New CA," the browser will navigate to a page showing the "HSM Provisioning Certificate." This will be used as a template to configure the new CA on the IoT-HSM appliance.

Note: If at any time you need to navigate back to this page, go to **Certificate Authorities -> Manage CAs**, click on the CA and select "Download HSM Provisioning Certificate" from the menu.

See the configuration documentation to configure IoT-HSM to sign requests for the new CA.

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14.4 Configuration

If you haven't yet created an HSM CA and retrieved the provisioning certificate, see the Create an HSM CA documentation before proceeding. Also, make sure your HSM is ready for use. See documentation for your HSM if it isn't showing up on the IoT-HSM dashboard.

- YubiKey
- Google Cloud KMS
- SoftHSM2 (Not recommended for production use)

Prepare the Private Key on the IoT-HSM

Log in to the IoT-HSM. From the "HSM Options..." dropdown on one of the available HSMs, choose "Provision a New Slot." This will navigate to the provisioning wizard. Choose the slot to provision and a key option.

For Google Cloud KMS keys, no additional options are needed because keys are managed within Google Cloud.

The "Generate" option (recommended) will generate a new private key on the HSM slot using the "Key Type" option provided. The "Import" option allows an existing private key to be imported to the HSM slot. This option should only be used when the private key has been handled with extreme care and is ideal under circumstances when it is important to have a backup of the private key or if multiple IoT-HSMs will be provisioned with the same private key for redundancy.

Import the HSM Provisioning Certificate to the IoT-HSM

SELF-SIGNED ROOT CA

After preparing the private key, the next page in the provisioning wizard is where the "HSM Provisioning Certificate" generated in a previous step will be submitted. If a self-signed root CA is being generated, leave the first option selected, paste the provisioning certificate into the text area, and click submit.

INTERMEDIATE CA

If this CA will be an intermediate CA, select the option to generate a CSR, paste the provisioning certificate into the text area, and click submit. A modal will appear with a CSR that will need to be signed by the root CA issuing the certificate for this intermediate. Finally, select the last option in the dropdown and paste the certificate signed by the root using the provided CSR and click submit.

IMPORT A SIGNED CERTIFICATE

Import a certificate after signing the CSR generated using the "Intermediate CA" option. This option may also be used after updating IoT-HSM or migrating to a new appliance and the original certificate needs to be re-associated with the key.

Note: If at any time you need to navigate back to this page, go to the IoT-HSM home page, select the slot being provisioned from the "Configured Slots..." dropdown, and click "Complete Provisioning."

Import the New Certificate from the IoT-HSM to PKIaaS.io

This is the final important step that will associate the newly provisioned HSM with the CA on PKIaaS.io.

Navigate to the IoT-HSM home page. Select the newly provisioned slot from the "Configured Slots..." dropdown and click "View Certificate." Copy the certificate to the clipboard and on PKIaaS.io, navigate to **Certificate Authorities -> Manage CAs**, click on the CA being provisioned, and select "Import Signed Certificate." Paste the certificate copied from the IoT-HSM into the text area and click "Import Certificate."

A prompt will appear instructing to restart the IoT-HSM device. On the IoT-HSM home page, click "Reload" in the navigation bar at the top. Click "Import Certificate" again on the PKIaaS.io import signed CA page. If the imported certificate is an intermediate CA and PKIaaS.io is unable to download the certificate chain, a modal will appear prompting for the chain, including all intermediates and the root, to be supplied before the import can be completed.

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Provisioning Complete

When the steps have all been completed successfully, all signing requests for the provisioned CA will be delegated to the HSM on the IoT-HSM. Signing operations will fail if the IoT-HSM is not online with the HSM configured/inserted.

The number of successfully connected HSM slots will appear at the top right of the IoT-HSM home page. It may be necessary to refresh the page to see the updated status, as it may take a few moments to connect to PKIaaS.io. When a slot is selected from the "Configured Slots..." dropdown on one of the HSMs, the status will be displayed, indicating whether the slot is connected successfully.

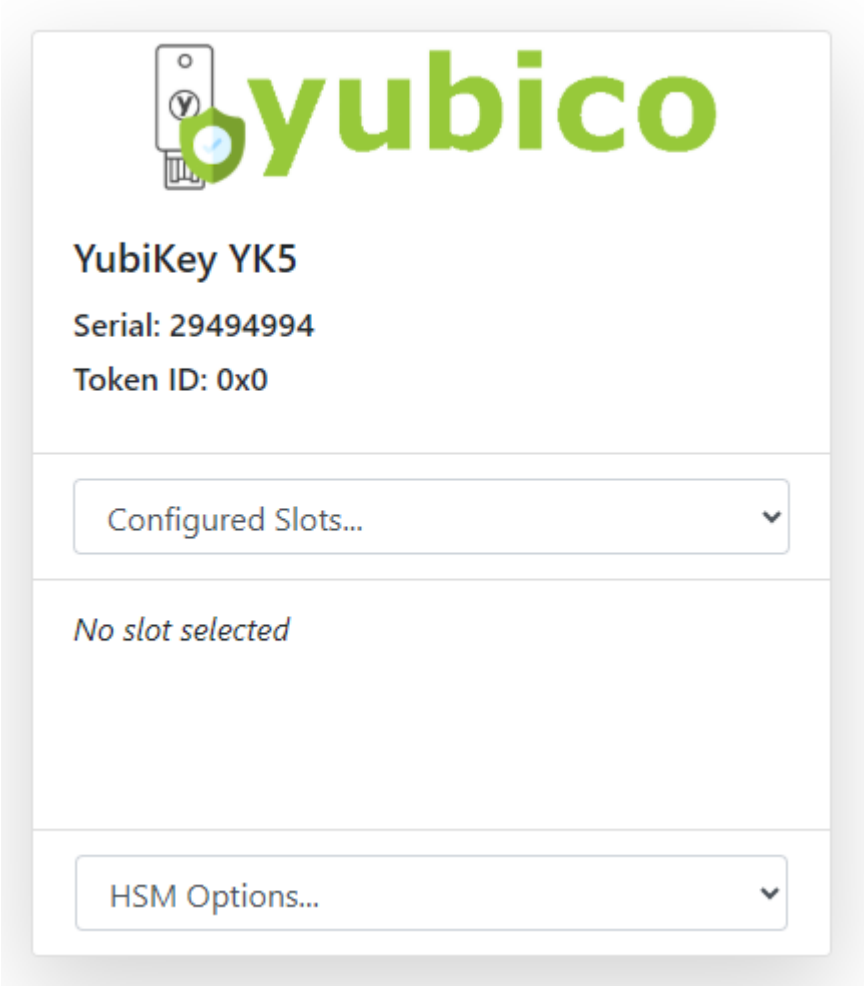
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14.5 Use a YubiKey

Insert the YubiKey into the IoT-HSM Appliance

All that is required to use a YubiKey as an HSM with IoT-HSM is to plug it into the appliance. Next, log in to the IoT-HSM. If you don't see the YubiKey listed on the main IoT-HSM page like the image below, use the reload button in the top navigation bar to refresh the list of HSMs.

Note: Multiple YubiKeys can be used on the same IoT-HSM appliance.



Next, see the configuration documentation to continue configuring the YubiKey with IoT-HSM.

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14.6 Use Google Cloud KMS

Log in to the Google Cloud Console

Navigate to the Google Cloud Console, and log in with your Google account.

Create a New Project and Add a Billing Account

If you haven't already, create a new project to contain the KMS key(s) that will be used with IoT-HSM. After the project is created, use the navigation menu at the top left to choose "Billing" and add a billing account.

Enable Required APIs

In the Google Cloud Console using the navigation menu at the top left, go to "APIs & Services" > "Enable APIs and Services". Ensure the following APIs are enabled for the project:

- Cloud Key Management Service (KMS) API
- Cloud Resource Manager API

Create a Key

In the Google Cloud Console, use the navigation menu to go to "Security" > "Key Management". Click on "Create key ring" to create a new key ring. Navigate to the new key ring and click "Create key" to create a new key within the key ring. Use the following options when creating keys:

- **Key name**: Key names may contain letters, numbers, underscores, and hyphens. They must be unique within the key ring and cannot be renamed or deleted.
- **Protection Level**: HSM
- **Key material**: HSM-generated key
- **Purpose**: Asymmetric sign
- **Algorithm**: Supported algorithms
- Elliptic Curve P-256 - SHA256 Digest
- Elliptic Curve P-384 - SHA384 Digest
- Elliptic Curve P-521 - SHA512 Digest
- 2048 bit RSA - PKCS#1 v1.5 padding - SHA256 Digest
- 3072 bit RSA - PKCS#1 v1.5 padding - SHA256 Digest
- 4096 bit RSA - PKCS#1 v1.5 padding - SHA256 Digest
- 4096 bit RSA - PKCS#1 v1.5 padding - SHA512 Digest
- **Key rotation**: Not available for asymmetric keys
- **Duration of 'scheduled destruction' state**: The number of days for which a key version remains in the "Scheduled for destruction" state after a request to destroy is submitted. During this time, users can restore the key. The value cannot be changed once the key is created. Default is 30 days

Note: IoT-HSM can manage multiple Google Cloud KMS keys, but the keys must exist on the same key ring because only one key ring may be configured per appliance. Additional appliances must be deployed to manage keys on more than one key ring.

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Create a Service Account

In the Google Cloud Console, use the navigation menu to go to "IAM & Admin" > "Service Accounts". Click "Create service account" and provide a name, service account ID, and optionally a description for the service account. After creating the service account, assign it the following roles:

- Cloud KMS Crypto Operator
- Cloud KMS CryptoKey Signer/Verifier
- Cloud KMS Viewer

If roles for the service account need to be updated, they can be managed by selecting the service account and then navigating to "Permissions" and clicking "Manage Access".

If IoT-HSM has been deployed as a VM or container on GCP, it can use the default service account associated with the VM or container for authentication. In this case, no service account key is needed. This is highly recommended as it simplifies the authentication process and reduces the risk of key exposure. Be sure to enable the "Cloud Platform" API within the access scope configuration on the VM. All documentation regarding service account keys is not applicable in this scenario.

Create Service Account Key

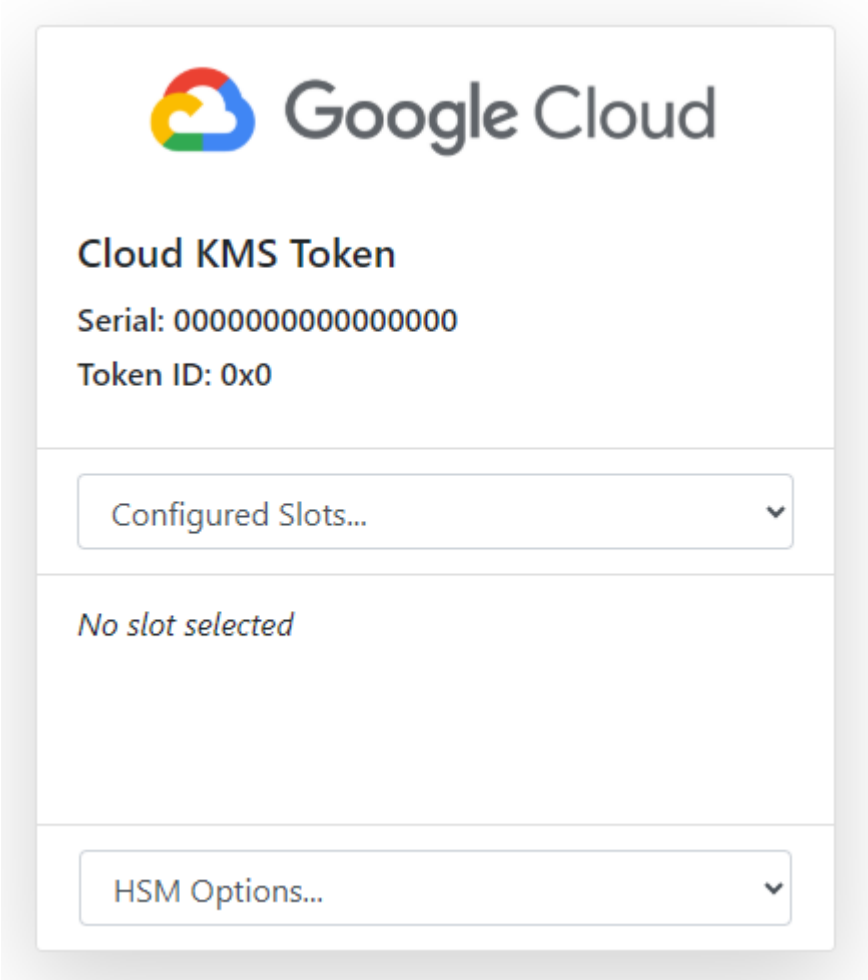
Select the service account you created earlier, then click "Keys" and "Add Key" to create a new key. Choose the JSON key type and click "Create". The key file will be downloaded automatically. Store this file securely, as it contains the credentials needed to authenticate as the service account.

Upload Service Account Key

Log into the IoT-HSM management interface and navigate to "Configure Google KMS". Upload the service account key file you downloaded earlier and click "Test Authentication". If authentication is successful, you will see the project and location fields populated with options to select. To select the appropriate key ring, choose the project and location it was created in. Finally, select the key ring to connect to the IoT-HSM and click "Check Permissions and Save".

Navigate to the IoT-HSM home page using the link at the top left and you should see a new "Cloud KMS Token".

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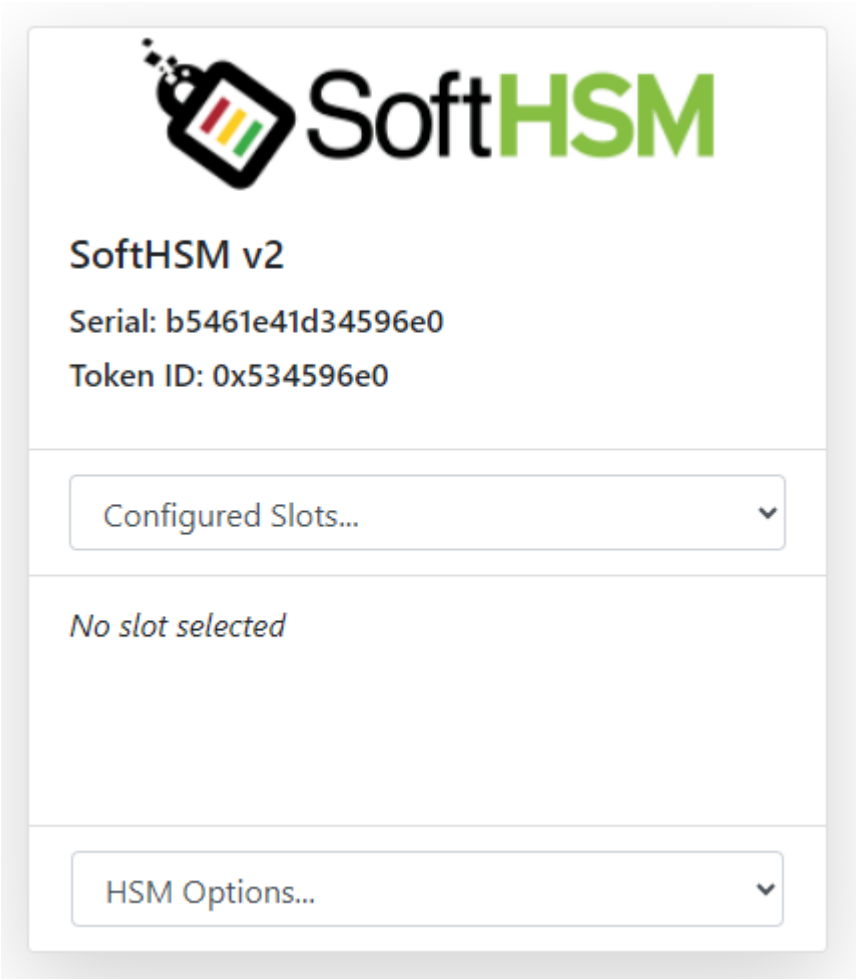
Next, see the configuration documentation to continue configuring Google Cloud KMS with IoT-HSM.

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14.7 Use SoftHSM2

Create a SoftHSM2 HSM

All that is required to create a SoftHSM2 slot is to navigate to the IoT-HSM in a browser and click "Create SoftHSM2" in the top left of the navigation bar. SoftHSM2 is a software-based HSM that provides a virtual device for cryptographic operations. It should be used only for testing IoT-HSM and is not intended for production use.



Next, see the configuration documentation to continue configuring SoftHSM2 with IoT-HSM.

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15. Troubleshooting

PKIaaS.io provides logging to help troubleshoot issues. Logs are available for backend services like SCEP and ACME to assist in diagnosing problems. These logs can be accessed by navigating to **Logs** in the navigation pane on the left side of the page. If you encounter issues with certificate issuance, SCEP, or other services, reviewing the logs can provide valuable insights into what went wrong. If further assistance is needed, you can reach out to the PKIaaS.io support team for help by direct message on Patreon or using the Disqus comment section on the help page of the website.

Note: Logs are retained for 30 days. After this period, they are automatically deleted to optimize storage and maintain system performance.

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